Market Incentives and the Drug Development Pipeline: Evidence from Antibiotics*

Edward Kong
Harvard University

Olivia Zhao
Harvard Business School

June 18, 2025

Abstract

Pharmaceutical innovation accounts for over one-third of recent life expectancy gains. This innovation is primarily incentivized by patents and FDA-awarded market exclusivities, but the effects of market exclusivity periods on different stages of the drug development pipeline are not well understood. In this paper, we estimate the causal effects of extending market exclusivity for antibiotics, a critical drug class where social value may exceed private returns. Using a difference-in-differences approach, we show that the Generating Antibiotic Incentives Now (GAIN) Act – which granted five additional years of exclusivity for antibiotics but not antivirals or vaccines – increased innovative activity across multiple stages of antibiotic development. Comparing antibiotics to a control group of antivirals and vaccines, we find positive effects on patenting, preclinical studies, and phase 3 clinical trials. We also find larger effects among drugs with more novel patents, and the majority of post-GAIN antibiotics contained novel active ingredients. These findings underscore the importance of financial incentives, such as market exclusivity extensions, in fostering high-quality innovation for socially valuable but commercially challenging drug categories.

*Contact: Edward Kong (edwardkong@fas.harvard.edu), Olivia Zhao (ozhao@hbs.edu). We are grateful for comments from Leila Agha, Amitabh Chandra, Rena Conti, Leemore Dafny, Anupam Jena, Yunan Ji, Jennifer Kao, Timothy Layton, Luca Maini, Kevin Outterson, and Mark Shepard. We also thank seminar participants at the Harvard Department of Economics, Harvard Medical School Department of Health Care Policy, the AcademyHealth Annual Research Meeting, the NBER Productivity Seminar, and the American Society of Health Economists (ASHEcon) Annual Meeting. Edward Kong received funding support from award Number T32AG51108 from the National Institute on Aging. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institute on Aging or the National Institutes of Health.

1 Introduction

Pharmaceutical innovation has led to significant decreases in mortality and morbidity (Dunn et al., 2022), accounting for over one-third of life expectancy gains between 1990 and 2015 (Buxbaum et al., 2020). However, the development of drug candidates into marketable products requires significant investments in R&D. Later stage development can involve hundreds of millions of dollars in clinical trial costs, which are primarily funded by private investment within pharmaceutical companies (DiMasi et al., 2016). To encourage innovation, private investment in drug development is primarily rewarded with statutory monopolies in the form of patents and market exclusivity periods, which trade off static inefficiencies from market power for long-term gains through innovation (Finkelstein, 2004). The resultant high drug prices have been the target of several important policies, such as the 2022 Inflation Reduction Act, which allowed the US government to negotiate prices for certain drugs for the first time. Such policies may achieve significant cost savings but may also reduce the flow of new welfare-improving drugs (Ho and Pakes, 2024; Adams, 2021).

Beyond the static-dynamic efficiency trade-off, another key policy issue is the potential mismatch between a given drug's private market returns and its social value. Private returns may correlate well with social value for some drug classes but poorly for others. Thus, current policies may simultaneously result in excessive R&D investment in some therapeutic areas (Mankiw and Whinston, 1986) but insufficient innovation in areas where private returns fall short of social value. In theory, tailoring market exclusivity periods based on social value could alleviate inefficiencies from private-social mismatch.

In this paper, we study whether strengthening market exclusivity incentives can bolster innovation across a rich set of different stages of the drug development pipeline, including patents, initial preclinical studies, clinical trials, and eventual drug launch. We focus on antibiotics as a key example of private-social returns mismatch, as evidenced by the paucity of new innovation in antibiotics despite rising antimicrobial resistance to existing antibiotics (Årdal et al., 2019; Outterson et al., 2015). Despite their medical importance, antibiotics have received relatively little attention within economics.

Our empirical analysis leverages the Generating Antibiotic Incentives Now (GAIN) Act, legislation created to address growing concerns over antibiotic resistance. The GAIN Act, first introduced in Congress in 2010 and enacted in 2012, extended the non-patent exclusivity period for certain "Qualifying Infectious Disease Products" (QIDPs) by an additional five years, a significant extension given the baseline exclusivity period of five years.[1] Non-

[1] This refers to the five years of exclusivity granted to New Chemical Entities (NCEs). Orphan Drugs are granted 7 years, whereas drugs containing a previously approved active ingredient may only obtain 3 years.

patent exclusivities granted by the U.S. Food and Drug Administration (FDA) represent an important policy lever and subject of study: while patent lengths have remained more or less constant over time, FDA exclusivities have been modified over time to achieve innovation objectives, such as increasing innovation in therapeutic areas with low private returns relative to social value (e.g., pediatric exclusivity and orphan drug exclusivity).

Identifying the causal effect of market incentives, such as exclusivity extensions, on innovation is challenging. Firms often choose whether to pursue these incentives based on factors that are not observable by the researcher, including drug candidates' unobserved quality. Additionally, candidates may be abandoned after obtaining poor clinical trial results, but abandonment is not easily observed in the data, and may only become clear after a drug fails to advance for several years. In our setting, firms may be more likely to seek QIDP designation for more promising drug candidates and are unlikely to seek QIDP designation for abandoned drug candidates.

We estimate the causal effects of the GAIN Act using a difference-in-differences approach, where the identification strategy hinges on the law's specific criteria for QIDP status. The GAIN Act only granted additional exclusivity to small-molecule antibiotic and antifungal drugs, but not other infectious disease products like antivirals and vaccines.[2] Accordingly, our analysis compares innovation before and after the GAIN Act for antibiotic and antifungal drug candidates (treated group) versus antivirals and vaccines (control group).

Identification relies on the parallel trends assumption, which is supported by the fact that our control indications are exposed to many of the same demand-side shocks that might affect market demand for antibiotics.[3] Crucially, however, antibiotic and antiviral drugs do not share similar capital and labor inputs; compounds targeting bacteria are unlikely to be effective against viruses, and vice versa. Moreover, firms in our data typically focus on either antibiotics or antivirals, but not both.

We use data from *Cortellis*, a comprehensive source that captures development milestones and related patenting activity for drugs in development. This allows us to estimate the GAIN Act's impact on patent applications, preclinical studies, and successful advancement at each stage of clinical development from phase 1 through phase 3 trials and US market launch.

Our ability to look at all stages of drug innovation is a major strength of this study for several reasons. First, examining earlier stages of innovation bolsters statistical power and may help us detect smaller innovation effects. One empirical challenge with studying innovation in general is that the arrival rate of new products may be low, which harms

[2]The eligibility restriction to small-molecule products implies that even vaccines targeting prevention of bacterial illnesses do not qualify for QIDP status.

[3]For example, viral respiratory illnesses are often followed by superimposed bacterial pneumonias.

statistical power. The prior literature has addressed this by examining innovation inputs (e.g., R&D spending) or intermediate measures of innovation (patents or patent citations). This challenge also applies in the pharmaceutical setting. The arrival rate of new drugs is low, because so few drug candidates surmount the winnowing process to become FDA approved. Moreover, the drug development process can take many years, from initial patenting through three phases of clinical trials to FDA approval. Analyzing changes in new drug approvals therefore requires many subsequent years of data, which inhibits researchers' ability to study the innovation impacts of new policies.

Fortunately, the pharmaceutical setting also offers a unique solution to the problem of infrequent innovation successes. The key insight is that all new drugs must proceed through stepwise clinical trials, and there are many more drug candidates in earlier phases of the drug development pipeline. As we show in Appendix A, phase-specific innovation increases imply increases in overall antibiotic approvals, as long as there are no offsetting phase-specific decreases. This is likely to be the case if the quality of marginal drugs is sufficiently high. Thus, analyzing phase-specific innovation effects yields greater statistical power by leveraging additional pipeline data. This also implies that an analysis that only examines FDA approvals leaves statistical power on the table, and will often fail to capture true innovation effects.

Second, market incentives may have heterogeneous effects on advancement at different stages of the innovation process. For a given (small) increase in incentives, the set of additional drugs now advancing to the next phase are those whose original profitability had just fallen short of their future development costs. The size of this marginal set of drugs may differ by development phase. There are several theoretical reasons for this. For example, the set of marginal drugs may be small between phase 1 and 2, because phase 1 trials primarily test safety in a small cohort of healthy patients. If the drugs that do not advance to phase 2 are simply those revealed to be unsafe, then market incentives are unlikely to matter at this stage.[4] Thus, estimating effect heterogeneity across clinical trial phases can provide information about the distribution of expected profits minus development costs at each phase. By including each phase in our analysis, we can potentially identify bottlenecks in the drug innovation pipeline that would respond better to policy interventions.

[4]The set of marginal drugs responding to market incentives may be larger between phases 2 and 3. Phase 2 trials reveal information about the expected efficacy (hence profitability) of a drug, and there may be a significant number of drugs whose expected profitability would not be enough to warrant conducting expensive phase 3 trials, but for the influence of market incentives. Another reason for potentially heterogeneous effects is that clinical trial costs vary significantly across phases. On one hand, if trial costs are small (as they are in earlier phases), then market incentives may not be pivotal for the decision of whether to conduct trials. On the other hand, if clinical trial costs are large (as they are for phase 3 trials), changes in expected profitability and financing frictions may matter much more.

Lastly, some drug candidates that "fail" in earlier stages may nonetheless prove to be important later on. A key example of this is the COVID-19 drug Paxlovid (nirmatrelvir/ritonavir), which was developed starting in 2020 based on an investigational compound for SARS created in 2003 that never made it to market.[5]

We find large effects of market incentives on some, but not all, stages of antibiotic innovation. First, we estimate a nearly 50% increase in patents for indications affected by the GAIN Act (antibiotics and antifungals) relative to control indications (antivirals and vaccines). Consistent with the language of the policy, increases were concentrated among QIDP-eligible patent types (i.e., new products, combinations, formulations, and processes), with no effect on QIDP-ineligible patent types (i.e., biologics). We also estimate a delayed, but statistically significant, 33% increase in preclinical studies by 2019, suggesting an eventual increase in the number of antibiotics entering the drug pipeline.

We then estimate how the GAIN Act affected the probability of advancing through each clinical trial phase through drug launch. We find a meaningfully large impact on the annual probability that a drug starting in phase 1 or 2 initiates a phase 3 clinical trial. We estimate that the GAIN Act increased phase 3 advancement rates by 4.9 percentage points, a 94% increase over the pre-period mean. Back-of-the-envelope calculations imply an expected revenue elasticity between 2.6 and 3.4, on the higher side of estimates from the existing literature. Consistent with our causal interpretation, we find larger innovation effects for drugs with older, but not expired patents; these drugs are more dependent on non-patent exclusivity to protect revenues. We also find that innovation increases are primarily driven by non-oral drugs (typically used for more serious infections) and pharmaceutical firms (versus public, academic, or hospital-based research groups).

Overall, examining multiple stages of development provides a more complete understanding of how the GAIN Act affected R&D for antibiotics and suggests that certain stages (e.g., patents and phase 3 trials) may respond more strongly to market incentives. That said, whether these intermediate measures of innovation will eventually result in additional drug approvals depends crucially on whether the marginal pharmaceutical innovations are valuable or novel (Agha et al., 2022; Dranove et al., 2022; Krieger et al., 2022). In the case of antibiotics, new drugs offer insurance against emerging antibiotic resistance, meaning that even new drugs with low realized demand can have high social value. We show two lines of evidence suggesting that the additional antibiotics were novel and valuable. First, we find larger clinical trial effects for drugs with product patents versus less novel patent types (e.g., new use, combination, or new formulation). Second, in a descriptive analysis of FDA-

[5]See https://cen.acs.org/pharmaceuticals/drug-discovery/How-Pfizer-scientists-transformed-an-old-drug-lead-into-a-COVID-19-antiviral/100/i3.

approved antibiotics, we show that antibiotics approved after the GAIN Act were more likely to contain novel active ingredients and less likely to be re-formulations or extended-release versions of older compounds.

Our results suggest that the GAIN Act partially reversed a decades-long decline in antibiotic innovation. We show that non-patent market exclusivity policies can be an effective tool for increasing innovation in areas where social benefits exceed private returns, especially in critical pharmaceutical classes with limited market potential. Finally, we highlight the practical utility of using additional data on intermediate stages of innovation to evaluate policy effects. This approach is useful in pharmaceuticals but could be applied to any setting where the arrival rate of new products is low or sporadic.

1.1 Related Literature

Our paper relates to an extensive prior literature on the effects of expanding expected market size or expected revenues on innovation (Acemoglu and Linn, 2004; Blume-Kohout and Sood, 2013; Dubois et al., 2015; Finkelstein, 2004). Other literature has focused on the effect of direct grants for R&D (Howell, 2017). Our paper extends this literature by studying a policy that extended market exclusivity instead of directly influencing revenues. This distinction is important, because policies that directly influence expected revenues, such as advance market commitments, remain rare (Kremer et al., 2020).

Currently, drug innovation is heavily reliant on patent and market exclusivities, leading naturally to the question: what is the optimal length of these exclusivities? New drugs benefit from exclusivity in two ways: patent protection and market exclusivity. Patent protection lasts for 20 years from the filing date, which usually precedes FDA approval. A growing body of work has highlighted the importance of patent exclusivity for pharmaceutical innovation (Budish et al., 2015; Gaessler and Wagner, 2018; Kyle and McGahan, 2012; Wang, 2022; Cockburn et al., 2016), although more research on intellectual property reform is needed (Bloom et al., 2019).

Market exclusivities, the topic of this paper, are directly awarded by the FDA for each drug product and induce a different set of firm incentives. While they run concurrently with patents, market exclusivities can outlast patent protection because they begin later (at the time of FDA approval). Unlike patents, market exclusivities cannot be challenged or nullified by the courts and are thus a lower risk way to protect drug revenue. An extended literature has explored strategic reasons for patenting, including creation of "patent thickets" to delay competitive entry (Galasso and Schankerman, 2010; Byrski and Wang, 2025). In contrast, firms cannot obtain additional market exclusivities for a given drug without running further

clinical trials.

For pharmaceuticals, the length of exclusivities has become a key policy lever. Market exclusivity extensions have been created for rare diseases, pediatric indications, and generic drugs. Most recently, the Inflation Reduction Act of 2022 allowed the Centers for Medicare and Medicaid Services to negotiate drug prices for high-revenue drugs starting 9 years after FDA approval for small-molecule drugs and 13 years after approval for biologics. This has resulted in significant spending decreases for negotiated drugs on the order of 22% according to recent estimates (Hernandez et al., 2024). Because these provisions effectively act like a cap on the exclusivity length, we can use our estimates to conduct a back-of-the-envelope calculation for the impact of the Inflation Reduction Act on pharmaceutical innovation.

Despite their unique importance, fewer studies have examined the impact of market exclusivities. An important early study by Yin (2008) showed that the Orphan Drug Act of 1983, which increased market exclusivity to 7 years for drugs treating rare diseases, increased innovation. Our study adds to this literature by studying market exclusivity in the new context of antibiotics, using updated drug pipeline data to assess the impact of exclusivities in a more recent scientific environment.

Beyond the setting of prescription drugs, our paper relates to classic questions within economics on innovation more generally. Given the importance of innovation across many industries, a longstanding question within economics has been the optimal design of intellectual property protection in terms of both patent length (Nordhaus, 1969; Scherer, 1972) and scope (Matutes et al., 1996). In the setting of pharmaceuticals, the FDA influences both length (via market exclusivities) and scope (via different exclusivity categories such as new molecular entities). Innovation effects are also important to many other industries where prices are wholly or partially regulated. For example, in the market for durable medical equipment, Medicare price reductions led to reductions in innovation that may fully offset savings (Ji and Rogers, 2024).

Our paper also contributes to the economics of antibiotics, an area where current innovation incentives are generally considered to be inadequate. Despite their importance, the economic issues surrounding antibiotics have received relatively limited attention in the field, with notable exceptions including Adda (2020) and Alsan et al. (2021). Understanding innovation incentives in therapeutic areas with tremendous social value and limited private value to innovators relates to antibiotics, as well as other neglected diseases that have been the subject of other policy interventions (Gans and Ridley, 2013; Ridley et al., 2021).

A closely related paper, Majewska (2022), features a structural model of push versus pull incentives for drug innovation. This paper also includes a reduced form analysis of the

GAIN Act using a different dataset (*Pharmaprojects*) and a different empirical approach.[6] Specifically, the treatment group in Majewska (2022) is constructed by combining antibiotic candidates *observed* to have obtained QIDP status plus antibiotic candidates with predicted QIDP status based on a random forest, with non-QIDP antibiotics serving as the control group. While this setup ensures that all drugs that eventually obtain QIDP status are included in the treatment group, it also assumes that attainment of QIDP status is uncorrelated with unobservables that may affect success rates.[7] In contrast, our method includes *all* antibiotics in the treatment group and does not condition on observed QIDP status. While not conditioning on observed QIDP status may help to address unobservables, it also means that our estimates should be considered as the effect of QIDP *eligibility* rather than the effect of actually acquiring QIDP status.

Like our paper, Majewska (2022) also finds large, positive effects of the GAIN Act on clinical trials.[8] We view our paper as providing complementary evidence for the effect of the GAIN Act by including evidence of extensive-margin effects on patents and preclinical studies, estimates specific to individual clinical trial phases, and evidence of heterogeneity by patent types and innovator characteristics (i.e., pharmaceutical firms vs other entities).

2 Background

The social value of antibiotics: In addition to providing the basis for our identification strategy, the setting of antibiotics is important in and of itself. In 1900, prior to the advent of antibiotics, infections caused 46% of *all* deaths (Armstrong et al., 1999; Gordon, 1953). By 1996, infectious disease mortality had dropped by 93%, due in part to the discovery and use of penicillin and sulfa antibiotics (Alsan et al., 2021; Armstrong et al., 1999; Jayachandran et al., 2010).

Antibiotics have become a staple of modern medicine, but the rise of antibiotic resistance threatens a return to a pre-antibiotic era (Baker, 2015). Each year, antibiotic resistance is associated with 2.8 million infections and over 35,000 deaths in the US (US Centers for Disease Control and Prevention, 2019) and 1.27 million deaths globally (Murray et al., 2022).

[6] Majewska (2022) estimates a difference-in-differences specification where the outcome variable is whether a candidate i in phase k *ever* advances to any later stage. In contrast, we examined yearly advancement rates.

[7] In addition, pre-period milestones are dropped for observed QIDPs, such that the pre-period treatment-group is limited to drugs that either (1) became FDA-approved prior to 2012 or (2) halted development prior to 2012. This could lead to estimated effects appearing smaller (in case 1) or larger (case 2). Our setup allows observations to begin development in the pre-period and continue development in the post-period.

[8] Majewska (2022) estimates that the GAIN Act increased the probability of preclinical projects entering clinical trials by 2021 by 31.5 percentage points and the probability of drugs in clinical trials advancing to a later phase by 21 percentage points, although this latter effect was not statistically significant.

Although many common bacteria are now resistant to penicillin and its immediate successors (Lobanovska and Pilla, 2017), the continuous invention of new antibiotics has preserved our ability to treat infections. Unfortunately, the rate of antibiotic innovation has stagnated in recent decades (Deak et al., 2016; Luepke et al., 2017; Wellcome Trust, 2020).

The simultaneous trends of increasing resistance and declining pharmaceutical investment raise the question of why the pharmaceutical industry develops so few new antibiotics despite high social need (Årdal et al., 2019; Outterson et al., 2015). One likely explanation is a lack of marketability for new antibiotics: novel antibiotics are often kept "in reserve," and many public health guidelines emphasize reductions in "inappropriate" antibiotic use (Centers for Disease Control and Prevention, 2014). These restrictions on use may limit the return on investment for antibiotics (Outterson et al., 2015).

Origins of the GAIN Act: The GAIN Act emerged against the backdrop of many large pharmaceutical firms terminating or withdrawing entire antibiotic pipelines (Koba, 2013; Plackett, 2020). After Pfizer shut down its main antibiotic research facility in 2011, the company's vice president of clinical research cited low financial returns from innovation as one of the decisive factors underlying the closure:

> "We were not having scientific success, there was no clear regulatory pathway forward, and the return on any innovation did not appear to be something that would support that program going forward."[9]

To increase incentives to develop new antibiotics, Congress introduced the GAIN Act in September 2010, immediately generating interest among infectious disease physicians (Infectious Diseases Society of America, 2011). The GAIN Act was signed into law in July 2012, representing one of the few legislative interventions passed in the US supporting antibiotic innovation.

The GAIN Act adds five additional years of market exclusivity for "Qualified Infectious Disease Products," defined as "an antibacterial or antifungal drug for human use intended to treat serious or life-threatening infections..." (U.S. Food and Drug Administration, 2018). The length of this additional exclusivity is significant, doubling the five-year exclusivity period for newly approved drugs (i.e., NCEs).[10] By comparison, the Orphan Drug Act only extended exclusivity by two years.[11] Despite its large effect on exclusivity, the GAIN Act's impact on innovation remains unclear (Darrow and Kesselheim, 2020).

[9] Dr. Charles Knirsch on FRONTLINE (Childress, 2013)

[10] Some drug products cannot qualify as NCEs and instead qualify for a shortened Clinical Investigation Exclusivity (CIE) period of 3 years based on conducting clinical trials for new indications, dosing regimens, or patient populations.

[11] Because patent protection also prevents entry from generic rivals, if drugs typically held at least one patent that expired after its market exclusivities, we would expect muted responses to policies that extend

Identification: The GAIN Act only affected antibiotics and antifungals, allowing us to use other infectious disease products (specifically, antivirals and vaccines) as a control group to identify the causal effect of additional market exclusivity. This choice of control group is a natural one, as both antibiotics and antivirals face similar demand shocks: see, for example, the co-occurrence of viral and bacterial pulmonary infections (Hament et al., 1999; Morris et al., 2017).

On the other hand, antibiotics and antivirals have distinct scientific development processes, suggesting minimal supply-side spillovers that could bias our estimates: firms are unlikely to conduct both antiviral and antibiotic R&D within a single research group or with the same scientists. In our sample of clinical trials, 87% of originating firms are associated with only treatment or only control group drugs, and this increases to 93% when conditioning on drugs observed to reach at least phase 2 clinical trials (i.e., drugs observed in phase 2 or phase 3 trials or US launch). That said, a handful of large pharmaceutical companies may have drug candidates in the treatment and control groups. Even weighting by firm portfolio size (i.e., the total number of treatment and control drugs by firm), 74% of drug candidates originate within a firm whose portfolio is at least 90% concentrated in either treatment or control drugs (79% when restricting to drugs with at least a phase 2 trial). This suggests that our results are unlikely to be driven by substitution between antiviral and antibiotic drug candidates within the same firm.

The GAIN Act therefore provides a unique, socially relevant setting for studying how exclusivity extension policies can affect drug innovation.

3 Methods

3.1 Data and Sample Definition

Our primary data source is Clarivate Analytics' *Cortellis* Competitive Intelligence database (Clarivate Analytics, 2022). *Cortellis* combines information from various public sources (e.g., press releases, financial filings, FDA submissions and approvals, clinical trial registries) to create a database of patents and drug pipelines with a timeline of the clinical trial development history for each drug. Importantly, the data contain non-clinical outcomes like patents and include drug candidates that never advanced beyond pre-clinical studies. Given the small number of antibiotics eventually approved by the FDA, the *Cortellis* data allow us to conduct a more well-powered analysis of innovation effects using patents and clinical trial

market exclusivity. That said, in the case of the Orphan Drug Act, the additional orphan drug exclusivity extended past the last expiring patent for 33 percent of qualifying drugs (Sarpatwari et al., 2018).

progression rates as outcome measures.

We systematically extracted the universe of drugs and patents in *Cortellis*. We restrict to patent filings and drugs with any development milestone occurring between 2005 and 2019, allowing us to observe six years of pre-GAIN trends and to exclude the COVID-19 pandemic beginning in 2020. We further restrict to observations with any associated indication under the "Infectious Disease" therapeutic area, which encompasses all bacterial, fungal, and viral infections.[12]

Cortellis contains other patent- and drug-level variables that we use in subsequent analyses, including firm originator name, associated patent identifiers and patent type classifications, and special regulatory designations (including QIDP status). To measure exclusivity periods for existing FDA-approved drugs, we use patent and market exclusivity data from the FDA Orange Book database (U.S. Food and Drug Administration, 1996).

3.2 Treatment Assignment

After extracting all drug candidates and patents within the Infectious Disease therapeutic area, we assign drugs to either the treatment or control group based on medical indications and vaccine/non-vaccine product type. Drug candidates and patents with antibiotic or antifungal indications (eligible for exclusivity extensions through GAIN) form our treatment group, while those with antiviral indications and antibacterial vaccines (non-eligible) form our control group.

We verified that our treatment group of antibiotics includes 110 of the 114 QIDPs in our dataset, compared to only 1 QIDP in our control group (Table 1).[13] This demonstrates that defining treatment and control based on indications captures nearly all drug candidates that eventually achieved QIDP status. By not explicitly conditioning on QIDP status, we avoid confounding from unobservable factors correlated with both receipt of QIDP status (which requires sponsors to provide some initial data demonstrating the drug's ability as an antibacterial or antifungal) and our outcomes. For example, drug candidates with unobservably high quality would be more likely to achieve QIDP status as well as reach clinical trial milestones.

[12] *Cortellis* classifies each indication attached to a patent or drug candidate into mutually exclusive and exhaustive therapeutic areas (e.g., the indication for "Strep Throat" falls within the Infectious Disease therapeutic area).

[13] There were 3 QIDPs that had both antibiotic and antiviral indications, which we excluded from our analysis. The single control group QIDP was for baloxavir marboxil, an influenza antiviral.

3.3 Key outcome variables

In *Cortellis*, we observe the exact dates that a drug candidate achieves the following outcomes, which capture innovative activity at different stages of development. Our key outcomes of interest, ordered chronologically from earlier to later in the drug development life cycle, are:

- *Newly filed patents: Cortellis* maintains a repository of patents linked to medical indications, allowing us to categorize patents into our treatment and control groups. *Cortellis* also classifies patents by type: "product" (referring to a new drug or therapeutic compound), "diagnostic" (related to methods or tests to diagnose diseases or conditions), and "ancillary" (a category we define to include combination, formulation, new use, and process patents). These categories allow us to assess whether certain types of inventions are more or less responsive to exclusivity incentives. We separately consider biologic patents (products or processes related to biologic drugs) in a falsification or "placebo" test, as biologics were explicitly exempt from the GAIN Act. We did not analyze transitions between patents and clinical trials, because reliable linkages between patents and individual drug candidates do not exist.

- *Pre-clinical activity ("phase 0"):* This stage represents new drug candidates that have entered into the first step of the development pipeline, but have not yet initiated human trials.[14]

- *Initiation of phase 1, phase 2, or phase 3 clinical trials:* Initiations of subsequent phases of clinical trials represent forward progress through the pipeline. Considering each of these phases separately allows us to identify which stage(s) are most responsive to exclusivity extensions and which stages appear to be bottlenecks. This information could inform policies targeted at specific stages. In addition, because clinical trials are initiated earlier and more frequently than drug launches (which may take many years), using trial initiations allows us to detect smaller, more rapid responses to new policies such as the GAIN Act.

- *Drug launches*: While drugs may be launched in separate countries at different times, we focus on US drug launches because GAIN Act incentives apply only to that market. Analyzing the share of phase 3 drugs launched each year allows us to test whether marginal drug candidates are less likely to achieve FDA approval.

[14] *Cortellis* contains 2 different designations for pre-clinical work: "pre-clinical" and "discovery." We combine these two categories into a single "pre-clinical" stage.

Drug candidates may accumulate multiple indications, but we count each candidate only once to avoid double counting the same drug. For any given milestone, we take the earliest date the candidates achieves the milestone across all indications. We also aggregate these outcomes to the yearly level, because milestones tend not to occur more often than once per year for each drug. Candidates with both treated and control indications are excluded from analysis.

3.4 Difference-in-Differences Specifications

Patents and Pre-clinical Studies: For our patent and pre-clinical study outcomes, we adopt the difference-in-differences specification shown below in Equation (1). This specification compares annual log counts of patents (or pre-clinical studies) for our treatment group of antibiotic/antifungal indications versus our control group of antiviral/vaccine indications.[15] We consider 2011 as the first treatment year, since the GAIN Act was introduced in Congress in 2010.[16]

$$\ln Y_{gt} = \sum_t \beta_t \times 1\{treated\}_g + \alpha_t + \alpha_{g,t<2011} + \mu_{gt} \tag{1}$$

The coefficients of interest β_t represent the difference in log patents or pre-clinical studies between treatment and control groups g in period t (where we normalize the average pre-period difference $\alpha_{g,t<2011}$ to zero); α_t represent year fixed effects, and μ_{gt} is the error term. We estimate the β_t coefficients by computing treatment-control differences $\ln Y_{treatment,t} - \ln Y_{control,t}$ for each year t and subtracting the average pre-period difference $\alpha_{g,t<2011}$. We obtain standard errors using a Bayesian bootstrap with 200 iterations (Van Kerm, 2022).[17]

Clinical Trial Phase Transitions: For our drug candidate panel dataset, our outcome variables Y_{it}^k capture whether drug candidate i progresses from development phase k to k' in year t. This specification, shown below in Equation 2, compares yearly trial progression

[15] As a robustness test, we also applied this specification to our phase transition outcomes described below (see Appendix E).

[16] The GAIN Act was originally introduced as H.R. 6331 on September 29, 2010, and later re-introduced as H.R. 2182 on June 15, 2011, and in the Senate as S. 1734 on October 19, 2011 (Generating Antibiotic Incentives Now Act, 2010, 2011*a*,*b*). Because firms' contemporaneous expectations included the probability that the bill would be passed in the future, we adopt 2011 as the first year that the GAIN Act may have had effects on firm expectations, despite the fact that the official signing of the bill occurred on July 9, 2012. The use of 2011 as our start year is consistent with the existence of policy briefs on the GAIN Act published by the Infectious Disease Society of America as early as February 15, 2011 (Infectious Diseases Society of America, 2011).

[17] Using the Bayesian bootstrap, as opposed to the standard parametric bootstrap, allows us to resample observations while avoiding draws of Y_{gt} containing zero observations, which would result in the logged value being undefined.

rates before versus after GAIN Act passage, between treatment and control drug candidates. Unlike with our patent and pre-clinical analyses described above, this allows us to control for drug-specific variables and drug fixed effects for greater statistical power. We estimate effects separately for each phase pair to capture potential heterogeneity. Our main specification focuses on phase 3 trials and measures the probability that a drug observed in phase 1 or phase 2 initiates a phase 3 trial.[18]

$$Y_{it}^{k,k'} = \sum_{t \neq 2010} \gamma_t \times 1\{treated\}_i + \delta_1 \times 1\{cohort\}_i + \delta_2 \times 1\{cohort\}_i \times 1\{treated\}_i + \alpha_t + \alpha_g + \varepsilon_{it}. \tag{2}$$

The coefficients of interest, denoted by γ_t, reflect how progression rates differ between treatment and control groups, relative to the reference year (2010). The coefficients δ_1 and δ_2 control for each drug's cohort[19], where we allow cohort effects to differ by treatment group. This captures group-specific differences in scientific progress at the time a drug candidate was first conceived[20] The α_t and α_g coefficients denote year and treated-group fixed effects respectively, and ε_{it} is the error term. We estimate Equation 2 using OLS with standard errors clustered at the drug level.

In the above specification, our identifying variation comes from pre-policy versus post-policy and treatment versus control comparisons *within each cohort.*[21] Trial progression may also differ by dosage form or whether a drug is a biologic or small molecule, so we also consider specifications that add these controls interacted with the treatment group. These specifications are more conservative compared to including drug fixed effects.[22]

Because our design only includes a single intervention year, our difference-in-difference estimates avoid recent concerns regarding heterogeneous treatment effects that apply to

[18]For each milestone k', we require that each drug candidate is observed to initiate a trial at the prior stage k; this ensures that the included drug candidates are truly "at risk" of initiating phase k'. For our main specification, we allow either phase 1 and phase 2 trials to satisfy this criterion, which helps account for any missingness in phase 1 trial reporting among drugs with observed phase 2 trials. This could occur, for example, if the drug was originally tested for a non-antibiotic indication in phase 1 and did not need to repeat phase 1 trials before further testing on antibiotic indications.

[19]We define a drug's cohort as the first year the drug is observed in our dataset at *any* phase, usually phase 0.

[20]The cohort controls also account for differences in data ascertainment. For example, if selective reporting of clinical trials in the earlier years of our sample favored successful candidates, then older drug cohorts would appear to have higher success rates.

[21]Cohorts starting after the GAIN Act help to pin down the year fixed effects but do not otherwise contribute to the identifying variation, which is limited to comparisons for cohorts and drugs under development in both pre- and post- periods.

[22]Using drug fixed effects produces results consistent with our baseline specification, with somewhat larger magnitudes (Figure 5.)

two-way fixed effects (TWFE) estimators (Baker et al., 2022; Roth et al., 2023).

Pooled Estimates: To compute an average treatment effect for the GAIN Act, we pool years within the pre- and post-periods, resulting in the following specifications shown in Equations (3) and (4), which parallel Equations (1) and (2). Here, the coefficients of interest $\tilde{\beta}$ and $\tilde{\gamma}$ represent average effects, comparing the post-period of 2011—2019 to the pre-period of 2005—2010.

$$\ln Y_{gt} = \tilde{\beta} \times 1\{treated\}_g \times Post_t + \tilde{\alpha}_t + \tilde{\alpha}_g + \tilde{\mu}_{gt} \tag{3}$$

$$Y_{it}^{k,k'} = \tilde{\gamma} \times 1\{treated\}_i \times Post_t + \tilde{\delta}_1 \times 1\{cohort\}_i + \tilde{\delta}_2 \times 1\{cohort\}_i \times 1\{treated\}_i + \tilde{\alpha}_t + \tilde{\alpha}_g + \tilde{\varepsilon}_{it} \tag{4}$$

Alternative Specifications Using Clinical Trial Counts: An alternative approach would be to use annual log counts of clinical trials at each phase, mirroring the specification we use for patenting and pre-clinical outcomes (Equation 1).[23] This is not our preferred specification for clinical trials because effects on earlier stages directly manifest in later stages. For example, a positive effect on the number of phase 3 trials could reflect an increase in transitions from phase 1 to 2 or from phase 2 to 3. Appendix Section E reports a robustness test using the log counts specification, with similar results.

Heterogeneity: We also explore treatment effect heterogeneity to support our identification strategy and elucidate mechanisms. For each indicator variable H_{it} encoding a dimension of heterogeneity (e.g., patent age), we estimate a modified version of Equation 4 where we interact all independent variables with H_{it}. We focus on dimensions of heterogeneity that are pre-determined or exogenous relative to clinical trial decisions, including patent type (considering product patents to represent more novel innovation compared to other patent types), patent age in a given year, dosage form (oral versus non-oral), and whether the originator is a pharmaceutical company (versus non-pharmaceutical entities such as universities). For patent type, we partitioned patents into four categories: product, diagnostic, biologic, and ancillary.[24]

[23] Log-transforming trial counts to express effects in percent terms is necessary given the large difference in group size between treatment and control drugs.

[24] The classification of patents into product, diagnostic, and biologic is defined and coded by *Cortellis*. The ancillary type combines the following *Cortellis* patent types: combination, formulation, new use, and process.

4 Results

4.1 Sample characteristics

Table 1 below shows descriptive statistics for our full sample, which contains N = 1,473 drug candidates in the treatment group and N = 3,485 drug candidates in the control group.[25] The treatment and control groups are reasonably well balanced on most characteristics except biologic status (with greater representation of biologics in the control group)[26]. Appendix Section H reports overall stage-specific success probabilities (e.g., the probability that a phase 1 drug initiates phase 2 trials) and discusses how our estimates compare to those in the existing literature. Generally speaking, launch rates are higher for antibiotics, because a greater share of antibiotics are launched after phase 3 trials. Reassuringly, the share of preclinical drugs reaching phase 3 trials is similar across both groups: 7.9% for treatment drugs versus 7.3% for control drugs.

Table 1. Descriptive Statistics

	Treatment	Control	Total
Total patent applications	26,589	27,157	53,746
Total unique drug candidates	1,473	3,485	4,958
Drugs with QIDP status	110	1	111
Oral dosage form	20%	16%	17%
Has matched patent	43%	41%	42%
Has product patent	18%	18%	18%
Private firm originator	81%	81%	81%
Orphan drug	9%	7%	7%

Notes: Table shows descriptive statistics for the full drug-level sample from 2005–2019. Treatment drugs are those with antibiotics or antifungal indications, control drugs include drugs with antiviral indications and antibacterial vaccines. Columns show statistics for treatment, control, and all drug candidates.

4.2 Effects on Patent Applications

This section reports results for patent applications, estimated using Equation (1). Patents have often been used as a proxy for innovation, as firms can only receive patents for "useful, novel, and non-obvious" inventions (Nagaoka et al., 2010). On one hand, increased patenting may reflect more investment in basic science research on antibiotics, which could lead to

[25] This is after dropping drugs that can be classified into both groups (76 drugs) and cohorts prior to 1995 (174 drugs).

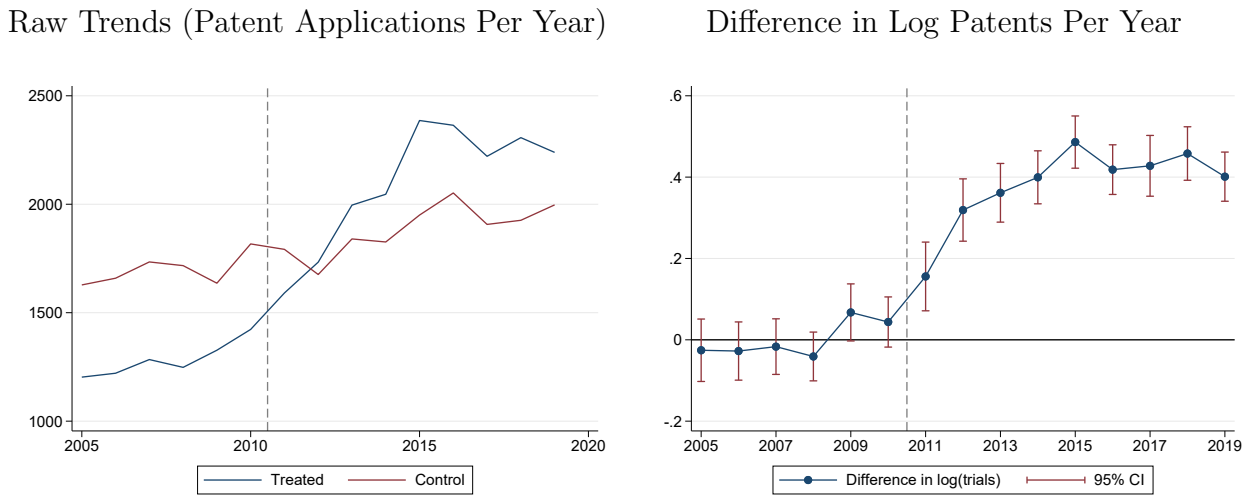
[26] Much of this difference is mechanical due assignment of vaccine products (all biologics) into the control group.

more new drugs in the long run. On the other hand, firms may simply be patenting a higher fraction of existing ideas. Either way, an increase in antibiotic patents would reflect a higher expected profitability of antibiotics, which would likely induce additional R&D investment in the long run.

We find that the introduction of the GAIN Act is associated with a significant rise in antibiotic patents compared to control patents. Figure 1 compares the log number of newly filed patents between the treatment versus control group from 2005 through 2019. Panel (a) plots raw trends in the number of treatment versus control patents. While the treatment group's pre-GAIN patenting level started below that of the control group, it surpassed the control group by 2012, representing a 47% increase (Appendix Table A4, Column 1).

Figure 1, Panel (b) plots differences in patenting activity between treatment and control, with confidence intervals obtained via Bayesian bootstrap. Reassuringly, there is no statistically significant pre-trend prior to the introduction of the GAIN Act. Immediately after the GAIN Act is introduced, we observe a sharp increase in antibiotic (treatment group) patents relative to the control group. While the raw trends in Panel (a) show a slight increase in antibiotic patents beginning around two years prior to GAIN enactment, the divergence between treatment and control becomes most clear starting the first year of the GAIN Act (2011).

Figure 1. Event Study of the GAIN Act on Patenting Activity



Notes: Figure shows the effect of the GAIN Act on the number of patent applications per year for treatment versus control indications as estimated through Equation (1). Panel (a) shows raw trends in the number of patent applications per year for both treatment and control groups. Panel (b) plots the difference in the log number of patents per year between treatment and control groups, with pre-period averages normalized to zero and confidence intervals obtained via Bayesian bootstrap. The timing of GAIN Act introduction is indicated by the dashed vertical line.

4.2.1 Heterogeneity in Patenting Effects

We explored potential sources of heterogeneity by estimating Equation 1 by patent type (e.g., product, diagnostic) and patent owner (e.g., pharmaceutical firm versus academic institution).

Increases in product, diagnostic, and ancillary patents: Figure 2 plots estimates from Equation (1) for each patent type and illustrates that relative patenting activity did not uniformly increase across different patent types.

The GAIN Act led to a statistically significant increase in product patent applications. Figure 2, Panel (a) shows that the increase in product patents peaks in 2014 before declining and potentially leveling off. This suggests a stock of ideas that led to a short-run increase in product patents, with a smaller persistent effect on the annual rate of new product patents. In aggregate, antibiotic product patenting increased by 12% in the post-period relative to the pre-period (Appendix Table A3). The fact that product patents increased is important, as product patents are generally considered to be particularly innovative or novel.

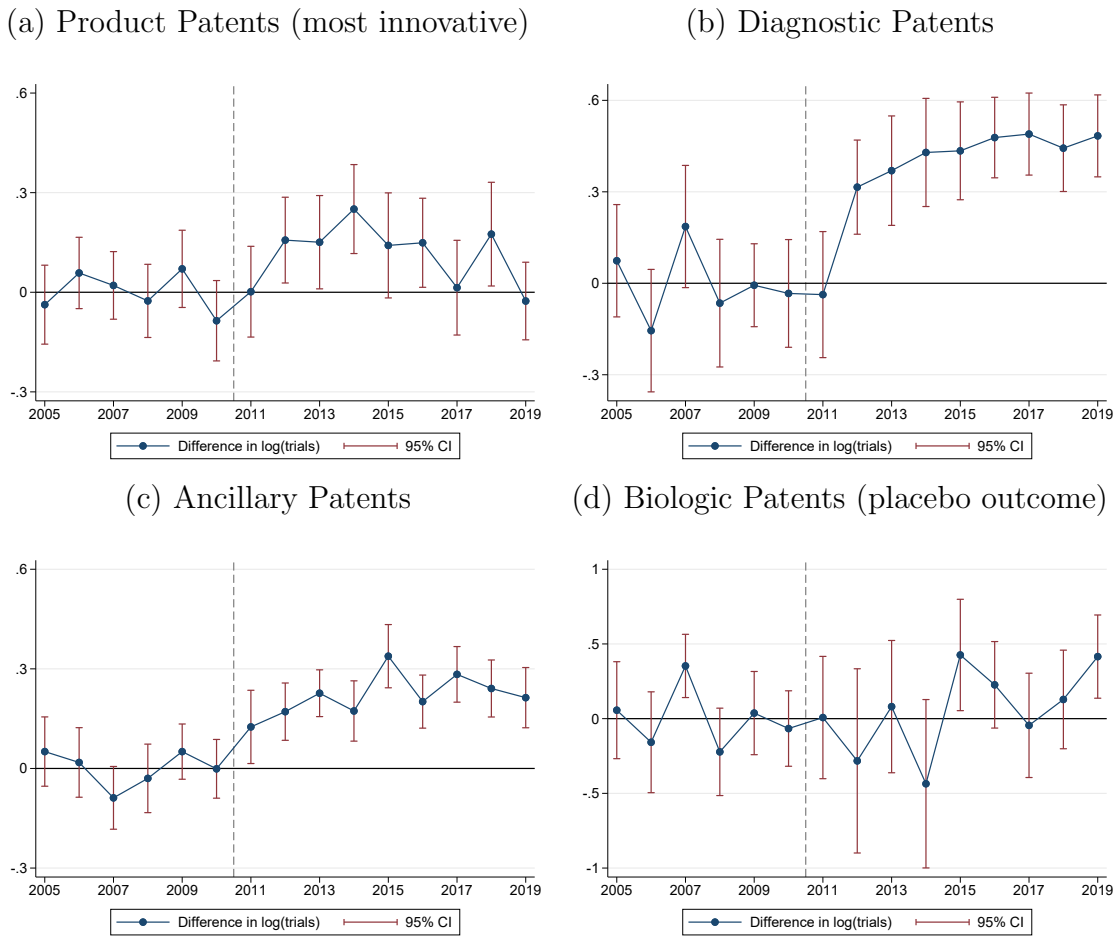
Diagnostics-related patents, which were QIDP-eligible under the GAIN Act, increased by 48%. Earlier versions of the GAIN Act (Generating Antibiotic Incentives Now Act, 2011*b*) explicitly included a provision to add 6 months of exclusivity for "companion diagnostic" tests, but this provision was not included in the version of the GAIN Act that was eventually passed (Food and Drug Administration Safety and Innovation Act, 2012). In 2021, FDA guidance clarified that the agency considers products intended to diagnose infections to be eligible for QIDP status, citing FDA precedent dating back to 1998 (U.S. Food and Drug Administration, 2021). Although we cannot determine the extent to which innovators believed that new diagnostics would directly benefit from the GAIN Act, our results are robust to the exclusion of diagnostic patents, which comprise a relatively small share of antibiotic patents overall (Appendix Table A3).

Even if firms did not believe that diagnostics could receive QIDP status, diagnostic patents could complement development or marketing efforts for antibiotic products. In terms of development, diagnostics can expedite clinical trials by improving screening and enrollment of patients into trials, potentially saving time and money by ensuring that only patients most likely to benefit from therapy are enrolled (Trevas et al., 2021). In terms of marketing, companion diagnostics could be used to drive demand for their associated antibiotic (Morel et al., 2016).

Ancillary patents exhibited a robust increase after the GAIN Act (Figure 2, Panel (c)), with an aggregate effect of 23% (Appendix Table A3, Column 4). Although typically considered less novel, these patents (which include combination, formulation, new use, and process

patents) can still be socially valuable. In the context of antibiotics, combination therapies are important for overcoming resistance.[27]

Figure 2. Heterogeneity in Patenting Effects by Patent Type



Notes: Each panel plots the difference in patenting between treatment and control indications for a different type of patent: product patents in Panel (a), diagnostic patents in Panel (b), ancillary patents (combination, formulation, new use, and process patents) in Panel (c), and biologic patents in Panel (d). Pre-period averages are normalized to zero and confidence intervals are obtained via Bayesian bootstrap. GAIN Act introduction is indicated by the dashed vertical line.

Biologic patents as a placebo outcome: Biologics are explicitly excluded from QIDP eligibility, as they are not approved under Section 505 of the FD&C Act (U.S. Food and Drug Administration, 2018). Thus, this category of patents serves as a placebo or falsification test. Reassuringly, we do not observe any effect of the GAIN Act on biologic patents (Figure

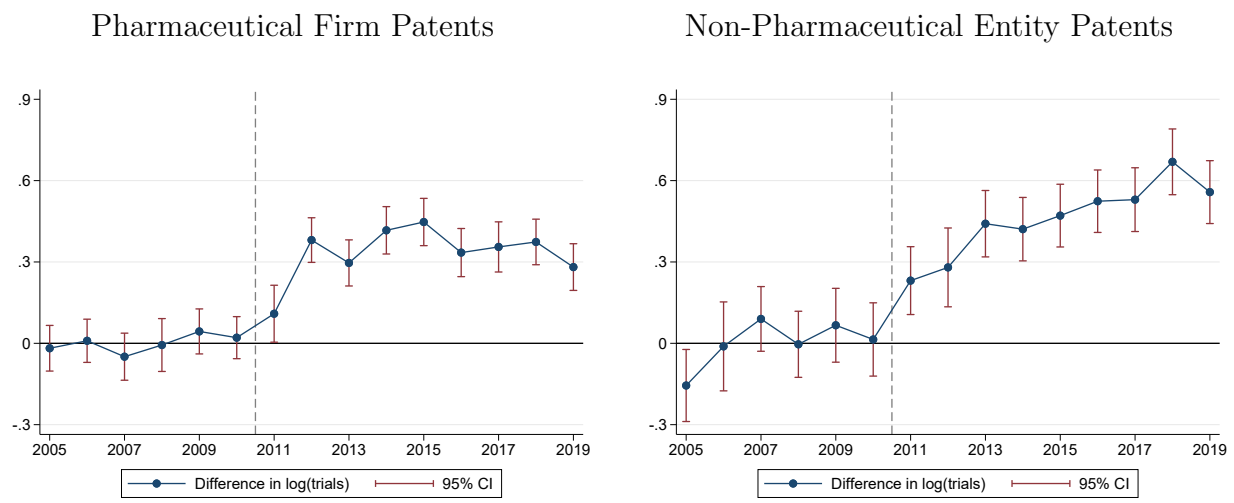
[27]Antibiotic combinations often take the form of an antibiotic combined with a compound that inhibits a resistant bacteria's evolved defenses against that antibiotic. One example of this is the antibiotic Avycaz (ceftazidime-avibactam).

2, Panel (d)). This lack of an effect on patents for an excluded subcategory of antibiotic drugs (biologics) lends strong support to our parallel trends assumption and choice of control group.

Heterogeneity by Pharmaceutical firms versus Non-Pharmaceutical entities: The majority of patents are owned by pharmaceutical firms, but around one-third are owned by non-pharmaceutical entities such as universities, hospitals, and non-profit institutes (e.g., the National Institutes of Health). Our data include 32,328 pharmaceutical patents versus 17,619 non-pharmaceutical patents.

We find that non-pharmaceutical owned patents exhibited a stronger *relative* increase in patenting compared to pharmaceutical/biotech company owned patents, 59% versus 41% (Appendix Table A4). This highlights the importance of public institutions in generating the basic research that feeds into later stages of drug development (Li et al., 2017). These results also suggest that economic incentives matter even for public entities, likely because subsequent development efforts (i.e., clinical trials) are driven by or in partnership with pharmaceutical firms. That said, because private firms conduct the bulk of patenting overall, private patents represent a larger share of the *absolute* increase in antibiotic patents induced by the GAIN Act.

Figure 3. Heterogeneity in Patenting Effects by Owner Type



Notes: Figures show the effect of the GAIN Act on the number of patent applications per year for treatment versus control indications estimated through Equation (1), separately for pharmaceutical owners (left) and non-pharmaceutical entities including universities, hospitals, and non-profit institutes (right). Pre-period averages are normalized to zero and confidence intervals are obtained via Bayesian bootstrap. The timing of GAIN Act introduction is indicated by the dashed vertical line.

Of note, we find that product patents increased by 40% among non-pharmaceutical entities, but did not change among pharmaceutical firms (Appendix Table A5). The increase in

patents owned by pharmaceutical firms was driven by non-product patents. In Section 4.4.1, we show that pharmaceutical firms also played an important role in advancing existing drug candidates through phase 3 clinical trials.

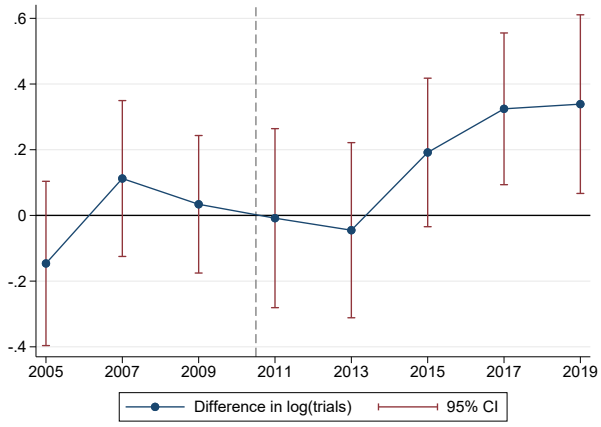
Taken together, our findings suggest that public entities took the lead on early-stage product innovation, which may later be acquired by larger firms with more clinical trials expertise. Meanwhile, pharmaceutical firms focused their efforts on non-product patents and advancing existing products through later development stages. This aligns with documented trends of public versus private innovation activities in the pharmaceutical industry (Cockburn and Henderson, n.d.).

Importantly, although we find evidence that public and private entities specialize on different stages of antibiotic innovation, both sectors responded to economic incentives. Put another way, public sector innovation is not completely immune to policies that primarily affect the private profitability of novel drugs.

4.3 Effects on Preclinical Studies

After patents, preclinical activity is the earliest indication that a firm is beginning development of a new drug product.Figure 4 below plots the difference in log preclinical studies between treatment versus control, estimated using Equation (1), with 95% confidence intervals obtained via Bayesian bootstrap. Pre-period trends are quite flat, lending support to the difference-in-differences parallel trends assumption. We observe a delayed uptick in new preclinical milestones beginning in 2015 and increasing to about +33% in 2019. The delay could be due to the time needed for sponsors to file patents if the new preclinical studies stem from discoveries covered by the new patents. However, due to the delay, the aggregate pre- versus post- comparison is not significant (Appendix Table A1).

Figure 4. Relative Difference in Preclinical Studies



Notes: Figure shows differences in log preclinical studies between treatment versus control groups estimated through Equation (1). Data are binned into two-year bins except for 2019, which contains one year. Pre-period averages are normalized to zero and confidence intervals are obtained via Bayesian bootstrap. The timing of GAIN Act introduction is indicated by the dashed vertical line.

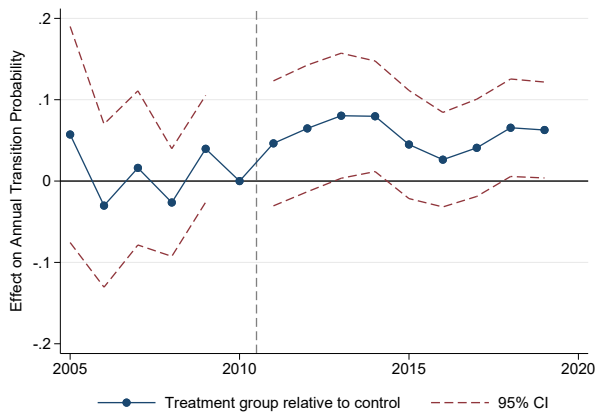
4.4 Effects on Progression Through Clinical Trials

As described in Section 3.3, Equation (2) allows us to estimate changes in transition probabilities through subsequent phases of clinical trials. Using success rates as opposed to the number of trial initiations accounts for the changing size of the pool of drug candidates that are "at-risk" for progressing to the next clinical trial phase. This allows us to infer whether level changes in observed clinical trial activity are due to higher rates of progression versus more candidates entering the pipeline in previous phases of development.

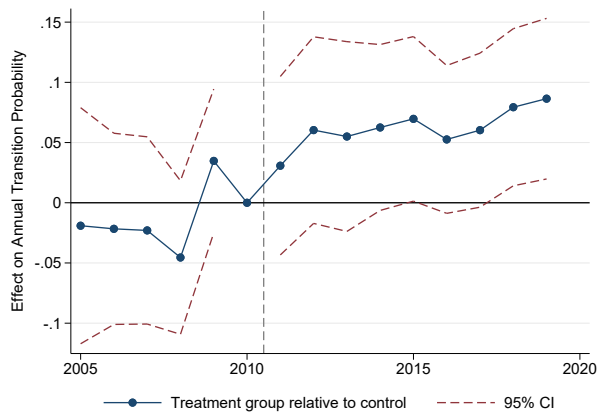
Figure 5 shows the results of the dynamic difference-in-difference specification in Equation (2) for the phase 1 to phase 3 transition, our main specification. The effects are modestly larger when controlling for drug fixed effects (Panel b) compared to the baseline specification that only controls for cohort-by-treatment fixed effects (Panel (a)). In both specifications, pre-period coefficients are not significantly different from zero, although our limited statistical power precludes a formal test for pre-trends. In Panel (a), progression rates for antibiotics from phase 1 to phase 3 begin to increase in 2011 and remain elevated throughout the post-policy period.

Figure 5. Effect on Advancement to Phase 3 Trials

(a) Baseline Specification



(b) Specification with Drug Fixed Effects



Notes: Figure shows the difference in phase 1 to 3 clinical trial advancement rates (the share of drugs that advance to phase 3, among drugs we observe in phase 1 or 2) between treatment and control groups, before versus after the GAIN Act was introduced in 2011 (indicated by the dashed vertical line). Panel (a) shows estimates from Equation (2) with treatment-by-cohort controls. Panel (b) shows estimates when drug fixed effects are included. Standard errors are clustered at the drug level.

Table 2 reports coefficients from the pooled specification in Equation 4 for the phase 1 to phase 3 transition. In our baseline specification (Column 1), we estimate that the GAIN Act increased phase 3 progression rates by 4.9 percentage points among drugs with phase 1 or 2 trials, a large effect of 4.9/5.2=94% of the pre-period mean. For pairs of adjacent phases, we find the largest effects between phase 2 and phase 3 (Appendix Figure A1 and Appendix Table A1).

These results suggest that the GAIN Act's effects on clinical trials are limited to the decision of whether to initiate phase 3 trials. This is plausible given the high costs of running phase 3 trials compared to earlier phases: phase 3 trials are by far the most expensive trials to run for infectious disease drugs, accounting for 55% of the average pre-approval clinical trial costs (Office of the Assistant Secretary for Planning and Evaluation, 2014). In contrast, phase 1 trials are small trials of a drug's inherent safety profile; therefore, we would not necessarily expect phase 1 to 2 transition rates to respond to financial incentives.

Table 2. Effect of the GAIN Act on Advancement to Phase 3 Clinical Trials

	Baseline	Dose X Treatment	Biologic X Treatment	Drug Fixed Effects
Post X Treatment	0.0486	0.0460	0.0469	0.0640
	(0.019)	(0.019)	(0.019)	(0.023)
Pre-period mean				
Treatment group	0.052	0.052	0.052	0.052
Control group	0.056	0.056	0.056	0.056
N unique drugs	1,336	1,336	1,336	1,133
N	7,403	7,403	7,403	7,200

Notes: Table shows pooled event study estimates from Equation (3) of the annual phase transition rate of drugs observed to enter phase 1 or 2 that initiate phase 3 trials. The coefficient of interest, Post X Treatment, reflects the effect of the GAIN Act on the transition rate of antibiotics (the treatment group) versus a control group comprised of antivirals and vaccines, which are not eligible for GAIN Act benefits. Column 1 shows our baseline specification, which includes fixed effects for the treatment group, year, cohort (the year a drug candidate is first observed in our dataset), and cohort X treatment (which adjusts for differences in cohort effects between treatment versus control groups). Column 2 also includes dosage form indicators and interactions with the treatment group. Column 3 adds a biologic indicator and interaction with the treatment group. Column 4 controls for drug fixed effects, where observations are dropped for drugs that are only observed once. Standard errors (in parentheses) are clustered at the drug level.

We do not find statistically significant changes in clinical trial activity for other phase transition pairs, and our confidence intervals allow us to rule out large effects (Appendix Table A1). For the phase 0 (preclinical) to phase 1 transition, our 95% confidence intervals rule out effects outside of -19% to +49% of the pre-period mean for the treatment group. For phase 1 to 2, we can rule out effects outside of -30% to +45%. This suggests that the increase in phase 3 trials was not offset by declines in phase 1 or 2 trials.

For drug candidates reaching phase 3, we do not find significant changes in the probability of launch. This suggests that the increase in new phase 3 candidates did not offset efforts to bring existing phase 3 candidates to market. Towards the end of our sample window, this also suggests that the additional phase 3 candidates induced by the GAIN Act were likely to be of similar quality to pre-existing phase 3 candidates. One caveat is that, given the small number of these drug candidates, we are only able to rule out effects on launch rates outside of -40 to +86% of the pre-period mean. We further explore the quality of new antibiotics introduced after the GAIN Act in Section 4.5 below.

In a specification aggregating all adjacent phase transition pairs, the average increase across all phases was +2.1pp (SE = 0.93pp, $P = 0.02$). This suggests that the overall effect of the GAIN Act on clinical trial progression was positive.

Robustness test using the log trials specification: Appendix Section E contains results for our alternative specification, where the outcome variable is defined as the log total number of trials in a given phase. This specification yields qualitatively similar results to the main results described above.

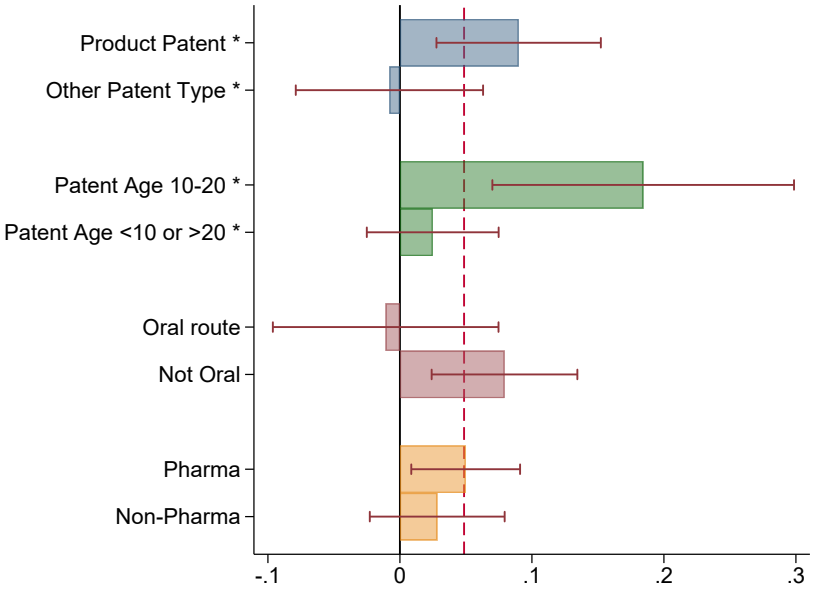
4.4.1 Effect Heterogeneity in Phase 3 Clinical Trial Advancement

To better assess the internal validity of our results and assess whether marginal innovations induced by the GAIN Act are truly "novel," we examine whether our estimated effects on phase 3 clinical trials differed across several drug characteristics, including aspects of linked patents (type and age), the drug delivery route (oral versus non-oral), and type of originator (pharmaceutical companies versus non-pharmaceutical entities). Although we previously examined heterogeneity in patenting effects, we now ask whether the GAIN Act's effect on subsequent clinical trial advancement differs by a drug candidate's underlying patent type and age.

Figure 6 shows estimates for heterogeneous treatment effects for the phase 1 to phase 3 transition.

***Effects on phase 3 clinical trials are larger for drugs with product patents*:** A major criticism of the GAIN Act and drug innovation policies more generally is that the marginal products may not represent novel or socially valuable innovations (Darrow and Kesselheim, 2020; Dranove et al., 2022). In contrast, we found that drug candidates matched to a product patent (which we consider to be more novel) had 9.8pp larger effects (SE=4.8pp) compared to those matched to new use, combination, reformulation, or process patents. This finding is consistent with our other descriptive evidence on the novelty of recently approved antibiotics (see Section 4.5). This finding is also consistent with the GAIN Act's stated focus on novel antibiotics. Note that while effects are larger for drugs with product patents, effects on product patents themselves (Section 4.2.1) are not as large as other patent types. This may reflect firms' desire to focus on novel products already in trials versus those at the patent stage, or it may reflect challenges with generating new product patents.

Figure 6. Effect on Advancement to Phase 3 Trials: Drug-Level Heterogeneity



Notes: Figure shows effect heterogeneity across drug characteristics for the phase 1 to phase 3 transition estimated through a versions of Equation (3) that interacts the difference-in-differences independent variables with each dimension of heterogeneity. The vertical dashed line represents the average treatment effect from our baseline specification without heterogeneity. Color-coded groups reflect binary characteristics: each bar represents the effect of the GAIN Act for a single group. Error bars represent 95% confidence intervals for each group-specific effect. The (*) indicates $p < 0.05$ for the between-group difference in treatment effects (i.e., the interaction term between the binary characteristic and Post X Treatment). Sample for "Product Patent" and "Patent Age 10-20" is restricted to observations with a matched patent. Sample for "Oral route" is restricted to observations with route of administration information.

Heterogeneity by patent age: Effects were entirely driven by drugs with patents between 10 and 20 years old (inclusive), suggesting an inverted-U shape relating patent age to the value of market exclusivity extensions.[28] We expected smaller responses for drugs with young patents (< 10 years old), as GAIN exclusivity extensions are less likely to extend beyond patent exclusivity expiration for these candidates.

The smaller effects for drugs with expired patents (> 20 years old) may initially be surprising, since these drugs would rely entirely on market exclusivity to limit direct competition. In actuality, this likely reflects a selection process: drugs that do not initiate phase 3 trials within 20 years of patent life likely have limited market potential or other negative attributes (e.g., lackluster phase 1 or phase 2 trial results). Our results suggest that this negative selection effect outweighs the mechanical increase in expected revenue from GAIN for drugs with expired patents.

[28] Only 4.5% of drugs with patent information have multiple patents listed. For these drugs, we select the oldest patent listed.

Overall, the larger effects we observe for drugs with older (but not expired) patents provide support for the internal validity of our research design.

Oral versus non-oral drugs: Our results are primarily driven by drug candidates with non-oral dosage forms, which are primarily intravenous (IV) medications.[29] Although the interaction term is not significant, the larger point estimate for IV drugs is consistent with the stated focus of GAIN that QIDP status should be awarded for "serious and life-threatening infections," which typically require intravenous antibiotics (Darrow and Kesselheim, 2020).

Pharmaceutical firms versus non-pharmaceutical entities: With the exit of some large pharmaceutical firms from antibiotics R&D, more attention has fallen to the role of non-pharmaceutical entities (universities, public research agencies like the National Institutes of Health, and hospitals) as the originating source of new antibiotic innovation. Interestingly, effects on phase 3 clinical trials appear to be driven primarily by traditional pharmaceutical firms, although we are not powered to detect significant differences between pharmaceutical firms and non-pharmaceutical entities.

Thus, it appears that pharmaceutical firms were an important contributor to the clinical trial response to the GAIN Act. That said, results from Section 4.2.1 show that patenting activity also increased substantially among non-pharmaceutical entities, albeit from a lower initial level. This points to an important role for non-pharmaceutical entities like universities in contributing to growth in early-stage innovation. However, non-pharmaceutical entities cannot entirely substitute for private firms, as pharmaceutical companies continue to account for the majority of both patenting and clinical trial activity.

4.5 Innovation Incentives and Drug Quality

Beyond the *number* of new antibiotics approved, policymakers also care about quality, in particular whether new antibiotics are novel, as novelty may correlate with social value in the antibiotics setting.[30] In Section 4.4.1 above, we showed that drugs with more novel product patents drive our GAIN Act effects. Here, we present a descriptive analysis that examines other measures of novelty, comparing the 67 antibiotics launched before and after GAIN during our study window. We abandon our control group for this analysis, since characteristics reflecting novelty in antibiotics (e.g., whether the antibiotic is a new dosage form of an old active ingredient) are not relevant for antivirals or vaccines.

For this descriptive exercise, we manually coded whether each new antibiotic includes

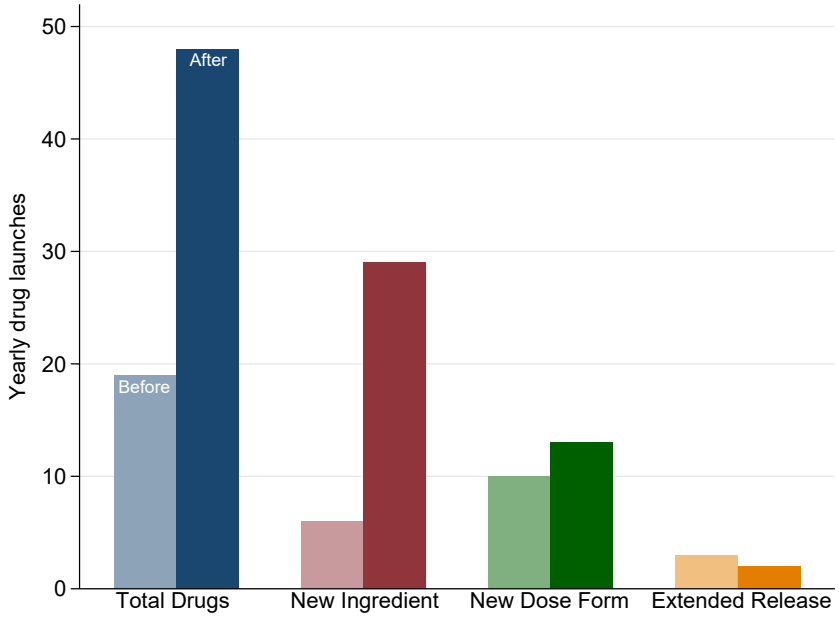
[29] Non-oral drugs also include other routes of administration, such as inhalation.

[30] Social value is also derived from policies that induce innovation in areas where social need is higher. In theory, all QIDPs should satisfy this criterion, as QIDPs are required to address "life-threatening conditions."

a novel active ingredient, a new dosage form or route of administration,[31] or was simply an extended-release version of an existing drug. One antibiotic (lefamulin) represented an entirely new class.

Figure 7 below shows that antibiotics launched after the GAIN Act appear somewhat *more* novel than antibiotics launched prior. After the GAIN Act, the launch rate of antibiotics increased (from 3.3 per year to 4.8 per year). A larger percentage of these drugs contained novel active ingredients (increasing from 30% to 60%), and a lower percentage represented new dosage forms (declining from 55% to 27%) or extended-release formulations of existing drugs (declining from 15% to 4%). In sum, after the GAIN Act, antibiotics launched at a faster rate, with a larger share featuring novel active ingredients.

Figure 7. Characteristics of New Antibiotics Before and After GAIN



Notes: Figure shows characteristics of N=67 treatment-group drugs that were first launched in the 6 years prior to GAIN (2005-2010) or after the GAIN Act was announced (2011-2020). The first bar in each group represents 2005-2010; the second bar in each group represents 2011-2020. The Y-axis reports the yearly rate of new drug launches in each category-by-time period. Group definitions are as follows: Total Drugs (yearly average number of all new antibiotic launches), New Ingredient (antibiotics with a novel active primary ingredient), New Dose Form (antibiotics with a previously approved active ingredient in a novel dosage form), and Extended Release (new antibiotics whose main novelty is an extended-release formulation). The sample of treatment group drugs also includes some antifungals and protein-based therapies, such as monoclonal antibodies.

These descriptive results align with our causal findings in Section 4.4.1, where product

[31] New dosage forms and delivery routes be valuable to treat specific clinical syndromes; for example, inhaled versus intravenous tobramycin for lung infections.

patents drove our difference-in-differences results. Both the difference-in-differences estimates and these descriptive facts help to highlight the conceptual distinction between innovation "quality" and marketability: many antibiotics that yield modest private returns may nonetheless be socially valuable.

5 Discussion

5.1 Interpreting the Magnitude of Our Estimated Effects

Our estimated effects are meaningfully large. Increasing market exclusivity for antibiotics by five years led to a 47% increase in patents, a significant increase in pre-clinical studies in later years (33% in 2019), and a 4.9 percentage point increase in the advancement rate from phase 1 to phase 3 clinical trials.

To compare our findings to prior estimates in the literature, we can translate our results into an elasticity of innovation with respect to revenue. For this exercise, we focus on our phase 1 to phase 3 clinical trial results.[32] We compute revenue elasticities in the range of 2.6 to 3.4 (Appendix Section B). These elasticity values are consistent with the prior literature on revenue elasticities of drug innovation, albeit on the larger end of the range. For comparison, Acemoglu and Linn (2004) find elasticities of about 4, Finkelstein (2004) finds elasticities of 2.75 for vaccines, and Dubois et al. (2015) finds an elasticity of 0.23, although this latter estimate is based on new molecular entities (NMEs), which are a subset of all new drugs. Comparing elasticities across these references is challenging as they span a range of therapeutic areas and time periods.

The limited private incentives to develop antibiotics prior to the GAIN Act may rationalize the larger magnitude of our estimates relative to other therapeutic areas. Antibiotics are unique in that recommended utilization practices, like holding new antibiotics "in reserve," may lead to artificially low private innovation incentives (Hamad, 2010). This lower baseline level of antibiotic innovation might mean that modest additional investment yields a sizable increase in innovative output.

5.2 Policy Implications

As an industrial policy for spurring drug innovation, market exclusivity extensions have several favorable characteristics. As opposed to innovation grants or prizes, exclusivity extensions preserve positive selection based on private willingness to pay: drugs with higher

[32] While this likely underestimates the total effect of the GAIN Act, it avoids assumptions about how to incorporate the earlier-stage effects of additional patents and pre-clinical studies.

commercial value are more likely to be developed (Weyl and Tirole, 2012). If private returns are positively correlated with societal value, then marginal innovations should also improve social welfare. For drug categories for which private and social values clearly diverge, variations in market exclusivity could be employed as a kind of "Pigouvian tax or subsidy" to correct for these differences. Moreover, exclusivity extensions do not require governments to raise funds or decide which drug candidates to reward (Dubois et al., 2022). These features distinguish market exclusivity extensions from other policies affecting expected profits (such as pricing regulation, subscription models,[33] and direct grants[34]).

While our results are specific to antibiotics, they could have broader implications for other policies that affect exclusivity length, such as the recently enacted Inflation Reduction Act of 2022. The IRA allowed the Centers for Medicare and Medicaid Services to negotiate drug prices for high-revenue drugs starting 9 years after FDA approval, effectively acting as a cap on the length of exclusivity.[35] Using our preclinical and phase 3 estimates, our back-of-the-envelope calculations suggest that the IRA may decrease long-run drug innovation by 1.3%, or approximately 1.05 new drugs every year (see Appendix Section G for details).

6 Conclusion

In this paper, we report causal estimates of how market exclusivity affects each stage of the drug innovation pipeline. Using a policy that extended antibiotic exclusivity by five years (the GAIN Act) and a variety of outcomes from patents to clinical trials, we provide a comprehensive view of how extending market exclusivity affected an important drug class. We find that the GAIN Act increased patenting, pre-clinical studies, and clinical trial progression for antibiotics.

Our results suggest that policies affecting the length of market exclusivity are likely to have significant effects on innovation. That said, over a decade after the GAIN Act's passage, antibiotics are still considered unprofitable, as evidenced by the ongoing bankruptcies of antibiotic developers (Mosbergen, 2023). The challenge of rising antibiotic resistance highlights the importance of further research into how market incentives affect the pipeline for new antibiotics.

Beyond the specific context of antibiotics, our results suggest that market exclusivity policies can increase innovation in other therapeutic areas where social benefits may exceed

[33]Examples include subscription payments by England's National Health System in 2019 and the PASTEUR Act proposed in the US (Doyle, 2020; PASTEUR Act, 2023).

[34]For example, CARB-X, a non-profit funded by various international governments and foundations, selects a group of antibiotic candidates to fund directly.

[35]The 9-year threshold applies to small molecule drugs. The threshold for biologic drugs is 13 years.

private returns. Further research is needed on the innovation effects of other policies, such as price regulation, market entry rewards, or subscription models. Careful attention to how public policies affect financial incentives for drug development will be crucial for determining whether society obtains essential new drugs, including antibiotics, in the future.

References

Acemoglu, Daron, and Joshua Linn. 2004. "Market Size in Innovation: Theory and Evidence from the Pharmaceutical Industry." *The Quarterly Journal of Economics*, 119(3): 1049–1090.

Adams, Christopher P. 2021. "CBO's Simulation Model of New Drug Development."

Adda, Jérôme. 2020. "Preventing the Spread of Antibiotic Resistance." *AEA Papers and Proceedings*, 110: 255–259.

Agha, Leila, Soomi Kim, and Danielle Li. 2022. "Insurance Design and Pharmaceutical Innovation." *American Economic Review: Insights*, 4(2): 191–208.

Alsan, Marcella, Vincenzo Atella, Jay Bhattacharya, Valentina Conti, Iván Mejía-Guevara, and Grant Miller. 2021. "Technological Progress and Health Convergence: The Case of Penicillin in Postwar Italy." *Demography*, 58(4): 1473–1498.

Årdal, Christine, Manica Balasegaram, Ramanan Laxminarayan, David McAdams, Kevin Outterson, John H. Rex, and Nithima Sumpradit. 2019. "Antibiotic development — economic, regulatory and societal challenges." *Nature Reviews Microbiology*, 18(5): 267–274. Publisher: Nature Publishing Group.

Armstrong, Gregory L, Laura A Conn, and Robert W Pinner. 1999. "Trends in Infectious Disease Mortality in the United States During the 20th Century." *JAMA*, 281(1): 61–66.

Aryal, Gaurab, Federico Ciliberto, Leland E. Farmer, and Ekaterina Khmelnitskaya. 2023. "Valuing Pharmaceutical Drug Innovations." *SSRN*.

Baker, Andrew C., David F. Larcker, and Charles C. Y. Wang. 2022. "How much should we trust staggered difference-in-differences estimates?" *Journal of Financial Economics*, 144(2): 370–395.

Baker, Stephen. 2015. "A return to the pre-antimicrobial era? | Science." *Science*, 347(6226): 1064–1066.

Bloom, Nicholas, John Van Reenen, and Heidi Williams. 2019. "A Toolkit of Policies to Promote Innovation." *Journal of Economic Perspectives*, 33(3): 163–184.

Blume-Kohout, Margaret E., and Neeraj Sood. 2013. "Market size and innovation: Effects of Medicare Part D on pharmaceutical research and development." *Journal of Public Economics*, 97: 327–336.

Budish, Eric, Benjamin N. Roin, and Heidi Williams. 2015. "Do Firms Underinvest in Long-Term Research? Evidence from Cancer Clinical Trials." *American Economic Review*, 105(7): 2044–2085.

Buxbaum, Jason D., Michael E. Chernew, A. Mark Fendrick, and David M. Cutler. 2020. "Contributions Of Public Health, Pharmaceuticals, And Other Medical Care To US Life Expectancy Changes, 1990-2015." *Health Affairs*, 39(9): 1546–1556. Publisher: Health Affairs.

Byrski, Dennis, and Lucy Xiaolu Wang. 2025. "Marketing Authorization and Strategic Patenting: Evidence from Pharmaceuticals."

Centers for Disease Control and Prevention. 2014. "Report to the President on Combating Antibiotic Resistance."

Childress, Sarah. 2013. "Dr. Charles Knirsch: "These Are Not Ruthless Decisions"."

Clarivate Analytics. 2022. "Cortellis Competitive Intelligence - Drug Timeline and Success

Rates."

Cockburn, Iain M., and Rebecca M. Henderson. n.d.. "Absorptive Capacity, Coauthoring Behavior, and the Organization of Research in Drug Discovery." 46(2): 157–182. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/1467-6451.00067.

Cockburn, Iain M., Jean O. Lanjouw, and Mark Schankerman. 2016. "Patents and the Global Diffusion of New Drugs." *American Economic Review*, 106(1): 136–164.

Darrow, Jonathan J, and Aaron S Kesselheim. 2020. "Incentivizing Antibiotic Development: Why Isn't the Generating Antibiotic Incentives Now (GAIN) Act Working?" *Open Forum Infectious Diseases*, 7(1): ofaa001.

Deak, Dalia, Kevin Outterson, John H. Powers, and Aaron S. Kesselheim. 2016. "Progress in the Fight Against Multidrug-Resistant Bacteria? A Review of U.S. Food and Drug Administration–Approved Antibiotics, 2010–2015." *Annals of Internal Medicine*, 165(5): 363–372. Publisher: American College of Physicians.

DiMasi, J A, L Feldman, A Seckler, and A Wilson. 2010. "Trends in Risks Associated With New Drug Development: Success Rates for Investigational Drugs." *Clinical Pharmacology & Therapeutics*, 87(3): 272–277. _eprint: https://ascpt.onlinelibrary.wiley.com/doi/pdf/10.1038/clpt.2009.295.

DiMasi, Joseph A., Henry G. Grabowski, and Ronald W. Hansen. 2016. "Innovation in the pharmaceutical industry: New estimates of R&D costs." *Journal of Health Economics*, 47: 20–33.

DiMasi, Joseph A., Maria I. Florez, Stella Stergiopoulos, Yaritza Peña, Zachary Smith, Michael Wilkinson, and Kenneth A. Getz. 2020. "Development Times and Approval Success Rates for Drugs to Treat Infectious Diseases." *Clinical Pharmacology & Therapeutics*, 107(2): 324–332. _eprint: https://ascpt.onlinelibrary.wiley.com/doi/pdf/10.1002/cpt.1627.

Dowden, Helen, and Jamie Munro. 2019. "Trends in clinical success rates and therapeutic focus." *Nature Reviews Drug Discovery*, 18(7): 495–496. Bandiera_abtest: a Cg_type: From The Analyst's Couch Publisher: Nature Publishing Group Subject_term: Drug discovery.

Doyle, Michael F. 2020. "The PASTEUR Act."

Dranove, David, Craig Garthwaite, and Manuel Hermosilla. 2022. "Does consumer demand pull scientifically novel drug innovation?" *The RAND Journal of Economics*, 53(3): 590–638. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/1756-2171.12422.

Dubois, Pierre, Olivier de Mouzon, Fiona Scott-Morton, and Paul Seabright. 2015. "Market size and pharmaceutical innovation." *RAND Journal of Economics*, 46(4): 844–871.

Dubois, Pierre, Paul-Henri Moisson, and Jean Tirole. 2022. "The Economics of Transferable Patent Extensions."

Dunn, Abe, Anne Hall, and Seidu Dauda. 2022. "Are Medical Care Prices Still Declining? A Re-Examination Based on Cost-Effectiveness Studies." *Econometrica*, 90(2): 859–886. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA17635.

Finkelstein, Amy. 2004. "Static and Dynamic Effects of Health Policy: Evidence from the Vaccine Industry." *The Quarterly Journal of Economics*, 119(2): 527–564. Publisher: Oxford University Press.

Food and Drug Administration Safety and Innovation Act, S.3187 112th Congress. 2012.

Gaessler, Fabian, and Stefan Wagner. 2018. "Patents, Data Exclusivity, and the Development of New Drugs." *Academy of Management Proceedings*, 2018(1): 17812.

Galasso, Alberto, and Mark Schankerman. 2010. "Patent thickets, courts, and the market for innovation." *The RAND Journal of Economics*, 41(3): 472–503. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1756-2171.2010.00108.x.

Gans, Joshua S., and David B. Ridley. 2013. "Innovation Incentives under Transferable Fast-Track Regulatory Review." *The Journal of Industrial Economics*, 61(3): 789–816. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/joie.12026.

Generating Antibiotic Incentives Now Act, H.R.2182 112th Congress. 2011*a*.

Generating Antibiotic Incentives Now Act, H.R.6331 111th Congress. 2010.

Generating Antibiotic Incentives Now Act, S.1734 112th Congress. 2011*b*.

Gordon, Tavia. 1953. "Mortality in the United States, 1900-1950." *Public Health Reports*, 68(4): 441–444.

Hamad, Bashar. 2010. "The antibiotics market." *Nature Reviews Drug Discovery*, 9(9): 675–676. Publisher: Nature Publishing Group.

Hament, Jeanne-Marie, Jan L.L. Kimpen, Andre Fleer, and Tom F.W. Wolfs. 1999. "Respiratory viral infection predisposing for bacterial disease: a concise review." *FEMS Immunology & Medical Microbiology*, 26(3-4): 189–195.

Hay, Michael, David W. Thomas, John L. Craighead, Celia Economides, and Jesse Rosenthal. 2014. "Clinical development success rates for investigational drugs." *Nature Biotechnology*, 32(1): 40–51. Publisher: Nature Publishing Group.

Hernandez, Inmaculada, Olivier J. Wouters, Emma M. Cousin, Ayuri S. Kirihennedige, and Sean D. Sullivan. 2024. "Interpreting The First Round Of Maximum Fair Prices Negotiated By Medicare For Drugs."

Ho, Kate, and Ariel Pakes. 2024. "Policy Options for the Drug Pricing Conundrum."

Howell, Sabrina T. 2017. "Financing Innovation: Evidence from R&D Grants." *American Economic Review*, 107(4): 1136–1164.

Infectious Diseases Society of America. 2011. "Combating Antimicrobial Resistance: Policy Recommendations to Save Lives." *Clinical Infectious Diseases*, 52(suppl_5): S397–S428.

Jayachandran, Seema, Adriana Lleras-Muney, and Kimberly V Smith. 2010. "Modern Medicine and the Twentieth Century Decline in Mortality: Evidence on the Impact of Sulfa Drugs." *American Economic Journal: Applied Economics*, 2(2): 118–146.

Ji, Yunan, and Parker Rogers. 2024. "The Long-Run Impacts of Regulated Price Cuts: Evidence from Medicare."

Kaiser Family Foundation. 2023. "10 Prescription Drugs Accounted for $48 Billion in Medicare Part D Spending in 2021, or More Than One-Fifth of Part D Spending That Year."

Koba, Mark. 2013. "Big Pharma Exit: Who's Fighting the Superbugs?" Section: Health & Science.

Kremer, Michael, Jonathan Levin, and Christopher M. Snyder. 2020. "Advance Market Commitments: Insights from Theory and Experience." *AEA Papers and Proceedings*, 110: 269–273.

Krieger, Joshua, Danielle Li, and Dimitris Papanikolaou. 2022. "Missing Novelty in Drug Development." *The Review of Financial Studies*, 35(2): 636–679.

Kyle, Margaret K., and Anita M. McGahan. 2012. "Investments in Pharmaceuticals Before and After TRIPS." *The Review of Economics and Statistics*, 94(4): 1157–1172.

Li, Danielle, Pierre Azoulay, and Bhaven N. Sampat. 2017. "The applied value of public investments in biomedical research." *Science*, 356(6333): 78–81.

Lobanovska, Mariya, and Giulia Pilla. 2017. "Penicillin's Discovery and Antibiotic Resistance: Lessons for the Future?" *The Yale Journal of Biology and Medicine*, 90(1): 135–145.

Luepke, Katherine H., Katie J. Suda, Helen Boucher, Rene L. Russo, Michael W. Bonney, Timothy D. Hunt, and John F. Mohr III. 2017. "Past, Present, and Future of Antibacterial Economics: Increasing Bacterial Resistance, Limited Antibiotic Pipeline, and Societal Implications." *Pharmacotherapy: The Journal of Human Pharmacology and Drug Therapy*, 37(1): 71–84. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/phar.1868.

Majewska, Gosia. 2022. "Incentivizing novelty in antibiotic development."

Mankiw, N. Gregory, and Michael D. Whinston. 1986. "Free Entry and Social Inefficiency." *The RAND Journal of Economics*, 17(1): 48–58. Publisher: [RAND Corporation, Wiley].

Matutes, Carmen, Pierre Regibeau, and Katharine Rockett. 1996. "Optimal Patent Design and the Diffusion of Innovations." *The RAND Journal of Economics*, 27(1): 60–83. Publisher: [RAND Corporation, Wiley].

Morel, Chantal, Lindsay McClure, Suzanne Edwards, Victoria Goodfellow, Dale Sandberg, Joseph Thomas, and Elias Mossialos. 2016. "Co-development of antibiotics and diagnostics for bacterial infection." In *Ensuring innovation in diagnostics for bacterial infection: Implications for policy [Internet]*. European Observatory on Health Systems and Policies.

Morris, Denise E., David W. Cleary, and Stuart C. Clarke. 2017. "Secondary Bacterial Infections Associated with Influenza Pandemics." *Frontiers in Microbiology*, 8. Publisher: Frontiers.

Mosbergen, Dominique. 2023. "The World Needs New Antibiotics. The Problem Is, No One Can Make Them Profitably." *Wall Street Journal*.

Murray, Christopher J. L., Kevin Shunji Ikuta, Fablina Sharara, Lucien Swetschinski, Gisela Robles Aguilar, Authia Gray, Chieh Han, Catherine Bisignano, Puja Rao, Eve Wool, Sarah C. Johnson, Annie J. Browne, Michael Give Chipeta, Frederick Fell, Sean Hackett, Georgina Haines-Woodhouse, Bahar H. Kashef Hamadani, Emmanuelle A. P. Kumaran, Barney McManigal, Sureeruk Achalapong, Ramesh Agarwal, Samuel Akech, Samuel Albertson, John Amuasi, Jason Andrews, Aleskandr Aravkin, Elizabeth Ashley, François-Xavier Babin, Freddie Bailey, Stephen Baker, Buddha Basnyat, Adrie Bekker, Rose Bender, James A. Berkley, Adhisivam Bethou, Julia Bielicki, Suppawat Boonkasidecha, James Bukosia, Cristina Carvalheiro, Carlos Castañeda-Orjuela, Vilada Chansamouth, Suman Chaurasia, Sara Chiurchiù, Fazle Chowdhury, Rafai Clotaire Donatien, Aislinn J. Cook, Ben Cooper, Tim R. Cressey, Elia Criollo-Mora, Matthew Cunningham, Saffiatou

Darboe, Nicholas P. J. Day, Maia De Luca, Klara Dokova, Angela Dramowski, Susanna J. Dunachie, Thuy Duong Bich, Tim Eckmanns, Daniel Eibach, Amir Emami, Nicholas Feasey, Natasha Fisher-Pearson, Karen Forrest, Coralith Garcia, Denise Garrett, Petra Gastmeier, Ababi Zergaw Giref, Rachel Claire Greer, Vikas Gupta, Sebastian Haller, Andrea Haselbeck, Simon I. Hay, Marianne Holm, Susan Hopkins, Yingfen Hsia, Kenneth C. Iregbu, Jan Jacobs, Daniel Jarovsky, Fatemeh Javanmardi, Adam W. J. Jenney, Meera Khorana, Suwimon Khusuwan, Niranjan Kissoon, Elsa Kobeissi, Tomislav Kostyanev, Fiorella Krapp, Ralf Krumkamp, Ajay Kumar, Hmwe Hmwe Kyu, Cherry Lim, Kruy Lim, Direk Limmathurotsakul, Michael James Loftus, Miles Lunn, Jianing Ma, Anand Manoharan, Florian Marks, Jürgen May, Mayfong Mayxay, Neema Mturi, Tatiana Munera-Huertas, Patrick Musicha, Lilian A. Musila, Marisa Marcia Mussi-Pinhata, Ravi Narayan Naidu, Tomoka Nakamura, Ruchi Nanavati, Sushma Nangia, Paul Newton, Chanpheaktra Ngoun, Amanda Novotney, Davis Nwakanma, Christina W. Obiero, Theresa J. Ochoa, Antonio Olivas-Martinez, Piero Olliaro, Ednah Ooko, Edgar Ortiz-Brizuela, Pradthana Ounchanum, Gideok D. Pak, Jose Luis Paredes, Anton Yariv Peleg, Carlo Perrone, Thong Phe, Koukeo Phommasone, Nishad Plakkal, Alfredo Ponce-de Leon, Mathieu Raad, Tanusha Ramdin, Sayaphet Rattanavong, Amy Riddell, Tamalee Roberts, Julie Victoria Robotham, Anna Roca, Victor Daniel Rosenthal, Kristina E. Rudd, Neal Russell, Helio S. Sader, Weerawut Saengchan, Jesse Schnall, John Anthony Gerard Scott, Samroeng Seekaew, Mike Sharland, Madhusudhan Shivamallappa, Jose Sifuentes-Osornio, Andrew J. Simpson, Nicolas Steenkeste, Andrew James Stewardson, Temenuga Stoeva, Nidanuch Tasak, Areerat Thaiprakong, Guy Thwaites, Caroline Tigoi, Claudia Turner, Paul Turner, H. Rogier van Doorn, Sithembiso Velaphi, Avina Vongpradith, Manivanh Vongsouvath, Huong Vu, Timothy Walsh, Judd L. Walson, Seymour Waner, Tri Wangrangsimakul, Prapass Wannapinij, Teresa Wozniak, Tracey E. M. W. Young Sharma, Kalvin C. Yu, Peng Zheng, Benn Sartorius, Alan D. Lopez, Andy Stergachis, Catrin Moore, Christiane Dolecek, and Mohsen Naghavi. 2022. "Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis." *The Lancet*, 399(10325): 629–655. Publisher: Elsevier.

Nagaoka, Sadao, Kazuyuki Motohashi, and Akira Goto. 2010. "Chapter 25 - Patent Statistics as an Innovation Indicator." In *Handbook of the Economics of Innovation*. Vol. 2 of *Handbook of the Economics of Innovation, Volume 2*, , ed. Bronwyn H. Hall and Nathan Rosenberg, 1083–1127. North-Holland.

Nordhaus, William D. 1969. "An Economic Theory of Technological Change." *The American Economic Review*, 59(2): 18–28. Publisher: American Economic Association.

Office of the Assistant Secretary for Planning and Evaluation. 2014. "Examination of Clinical Trial Costs and Barriers for Drug Development." Office of the Assistant Secretary for Planning and Evaluation (Department of Health and Human Services).

Outterson, Kevin. 2021. "Estimating The Appropriate Size Of Global Pull Incentives For Antibacterial Medicines." *Health Affairs*, 40(11).

Outterson, Kevin, John H. Powers, Gregory W. Daniel, and Mark B. McClellan. 2015. "Repairing The Broken Market For Antibiotic Innovation." *Health Affairs*,

34(2): 277–285. Publisher: Health Affairs.

Parasrampuria, Sonal, and Stephen Murphy. 2022. "Trends in Prescription Drug Spending, 2016-2021." Assistant Secretary for Planning and Evaluation.

PASTEUR Act, S.1355 118th Congress. 2023.

Payne, David J., Michael N. Gwynn, David J. Holmes, and David L. Pompliano. 2007. "Drugs for bad bugs: confronting the challenges of antibacterial discovery." *Nature Reviews Drug Discovery*, , (6): 29–40.

Plackett, Benjamin. 2020. "Why big pharma has abandoned antibiotics." *Nature*, 586(7830): S50–S52. Bandiera_abtest: a Cg_type: Outlook Number: 7830 Publisher: Nature Publishing Group Subject_term: Economics, Drug discovery, Policy, Health care.

Ridley, David B., Pranav Ganapathy, and Hannah E. Kettler. 2021. "US Tropical Disease Priority Review Vouchers: Lessons In Promoting Drug Development And Access." *Health Affairs*, 40(8): 1243–1251. Publisher: Health Affairs.

Roth, Jonathan, Pedro H. C. Sant'Anna, Alyssa Bilinski, and John Poe. 2023. "What's trending in difference-in-differences? A synthesis of the recent econometrics literature." *Journal of Econometrics*, 235(2): 2218–2244.

Sarpatwari, Ameet, Reed F. Beall, Abdurrahman Abdurrob, Mengdong He, and Aaron S. Kesselheim. 2018. "Evaluating The Impact Of The Orphan Drug Act's Seven-Year Market Exclusivity Period." *Health Affairs (Project Hope)*, 37(5): 732–737.

Scherer, F. M. 1972. "Nordhaus' Theory of Optimal Patent Life: A Geometric Reinterpretation." *The American Economic Review*, 62(3): 422–427. Publisher: American Economic Association.

Stephens, Peter. 2015. "Stimulating Antibiotic R&D." *AMR Review*.

Thomas, David W, Justin Burns, John Audette, Adam Carroll, Corey Dow-Hygelund, and Michael Hay. 2016. "Clinical Development Success Rates 2006-2015." BIO Industry Analysis.

Trevas, Dana, Angela M Caliendo, Kimberly Hanson, Jaclyn Levy, Christine C Ginocchio, and for the Infectious Diseases Society of America. 2021. "Diagnostic Tests Can Stem the Threat of Antimicrobial Resistance: Infectious Disease Professionals Can Help." *Clinical Infectious Diseases*, 72(11): e893–e900.

US Centers for Disease Control and Prevention. 2019. "Antibiotic resistance threats in the United States, 2019." Centers for Disease Control and Prevention.

U.S. Food and Drug Administration. 1996. "Approved Drug Products with Therapeutic Equivalence Evaluations (Orange Book)."

U.S. Food and Drug Administration. 2018. "Qualified Infectious Disease Product Designation Questions and Answers Guidance for Industry." US Department of Health and Human Services Draft Guidance.

U.S. Food and Drug Administration. 2021. "Qualified Infectious Disease Product Designation." Publisher: FDA.

Van Kerm, Philippe. 2022. "EXBSAMPLE: Stata module to compute exchangeably weighted (or Bayesian) bootstraps." *Statistical Software Components*. Publisher: Boston College Department of Economics.

Vogel, Matthew, Pragya Kakani, Amitabh Chandra, and Rena M. Conti. 2024. "Medicare price negotiation and pharmaceutical innovation following the Inflation Reduction Act." *Nature Biotechnology*, 42(3): 406–412. Publisher: Nature Publishing Group.

Wang, Lucy Xiaolu. 2022. "Global drug diffusion and innovation with the medicines patent pool." *Journal of Health Economics*, 85: 102671.

Wellcome Trust. 2020. "The growing crisis in antibiotic R&D."

Weyl, E. Glen, and Jean Tirole. 2012. "Market Power Screens Willingness-to-Pay." *The Quarterly Journal of Economics*, 127(4): 1971–2003.

Wong, Chi Heem, Kien Wei Siah, and Andrew W Lo. 2019. "Estimation of clinical trial success rates and related parameters." *Biostatistics*, 20(2): 273–286.

Yin, Wesley. 2008. "Market incentives and pharmaceutical innovation." *Journal of Health Economics*, 27(4): 1060–1077.

Online Appendix
Market Exclusivity and Innovation: Evidence From Antibiotics

Edward Kong and Olivia Zhao

A Linking phase-specific effects to overall innovation

To translate our estimates of phase-specific impacts into a measure of overall innovation, we use a simple accounting framework that expresses total drug launches as the product of pipeline entry and phase transition probabilities. This approach allows us to quantify how observed changes at each stage contribute to aggregate innovation.

Let Y denote the total number of new antibiotics successfully launched. This can be expressed as the product of:

- N_0: number of drug candidates entering the pipeline (i.e., patents or preclinical studies),
- p_{01}: probability of transitioning from preclinical to Phase 1,
- p_{12}: probability of transitioning from Phase 1 to Phase 2,
- p_{23}: probability of transitioning from Phase 2 to Phase 3,
- p_{3L}: probability of transitioning from Phase 3 to launch.

Then total launches are given by:

$$Y = N_0 \cdot p_{01} \cdot p_{12} \cdot p_{23} \cdot p_{3L}$$

Taking the ratio of total antibiotic innovation after the policy Y_{post} to the total antibiotic innovation prior to the polic Y_{pre} gives:

$$\frac{Y_{\text{post}}}{Y_{\text{pre}}} = \frac{N_{0,\text{post}}}{N_{0,\text{pre}}} \cdot \frac{p_{01,\text{post}}}{p_{01,\text{pre}}} \cdot \frac{p_{12,\text{post}}}{p_{12,\text{pre}}} \cdot \frac{p_{23,\text{post}}}{p_{23,\text{pre}}} \cdot \frac{p_{3L,\text{post}}}{p_{3L,\text{pre}}}$$

Taking logs, we have:

$$\log\left(\frac{Y_{\text{post}}}{Y_{\text{pre}}}\right) = \log\left(\frac{N_{0,\text{post}}}{N_{0,\text{pre}}}\right) + \log\left(\frac{p_{01,\text{post}}}{p_{01,\text{pre}}}\right) + \log\left(\frac{p_{12,\text{post}}}{p_{12,\text{pre}}}\right) + \log\left(\frac{p_{23,\text{post}}}{p_{23,\text{pre}}}\right) + \log\left(\frac{p_{3L,\text{post}}}{p_{3L,\text{pre}}}\right)$$

The above equations make it clear that increases in the transition probability at individual phases, without offsetting changes to other transition probabilities, translate into increases in total innovation. For small changes x, we can use the approximation $\log(1 + X) \approx x$ to relate the growth in overall innovation to the sum of the growth rates of N_0 and each of the transition probabilities:

$$\text{Growth in } Y \approx \text{Growth in } N_0 + \text{Growth in } p_{01} + \text{Growth in } p_{12} + \text{Growth in } p_{23} + \text{Growth in } p_{3L}$$

In our data, we observe significant positive growth in N_0 (via patent and preclinical outcomes) and in transition probabilities p_{12} and p_{23}, while none of the transitions show significant decline. This implies an overall increase in Y, even if an analysis based on FDA approvals alone is underpowered.

B Revenue elasticity calculations

We use two approaches to approximate the average increase in exclusivity for antibiotics after GAIN. Both methods assume for simplicity that revenues arrive at a constant rate over the drug's exclusivity period and fall to zero after the drug goes generic.

The first (simpler) approach only considers GAIN's effect on market exclusivity for all antibiotics launched between 2005 and 2020. Using FDA Orange Book data, we calculate the increase in market exclusivity across all antibiotics, only some of which are approved with QIDP status and receive the GAIN exclusivity extension. This yields an increase in market exclusivity of 36% for the average antibiotic.

The second approach accounts for the fact that patent and market exclusivities run concurrently, and the total exclusivity of a drug is the maximum of the two types of exclusivities. Because the GAIN Act only increased market exclusivity, understanding its effect on total exclusivity requires accounting for the effect of patents. Below, we explain how accounting for patents decreases this increase to +28%.

Patents are valid for up to 20 years, and each antibiotic may have multiple patents. However, unlike market exclusivity, patent exclusivity can be challenged in court. However, patent challenges are difficult to predict. Thus, to assess the effect of GAIN on total (patent + market exclusivity), we use the empirical distribution of total exclusivity for antibiotics approved prior to the July 2012 (when the GAIN Act went into effect).

We define total exclusivity as the difference between the FDA approval date of the first generic and the FDA approval date of the branded antibiotic. Of 19 antibiotics approved prior to GAIN, we observe total exclusivity for 8 of them, which averaged 10.54 years, but with a large standard deviation of 5.76 years. To assess the effect of GAIN, we conducted a simulation to compare this pre-GAIN total exclusivity distribution with the post-GAIN market exclusivity distribution, which conferred an average of 10.0 years of market exclusivity (SD = 1.5 years). We simulated 10,000 draws from both distributions (assumed to be normally distributed with the above means and standard deviations) and applied the restrictions that (1) baseline exclusivity cannot be negative, and (2) GAIN can only add at most 5 years of additional exclusivity.

While the average post-GAIN market exclusivity is less than the average pre-GAIN total exclusivity (10.0 years versus 10.5 years), the large standard deviations mean that many drugs do benefit from additional GAIN exclusivity. Assuming that new drugs draw pre- and post-GAIN exclusivities independently, we find that 46% of drugs would benefit from GAIN exclusivity, with an average benefit of 3.3 years for these drugs. Including the 54% of drugs that would not benefit, we calculate that GAIN adds an average of 1.6 years of exclusivity

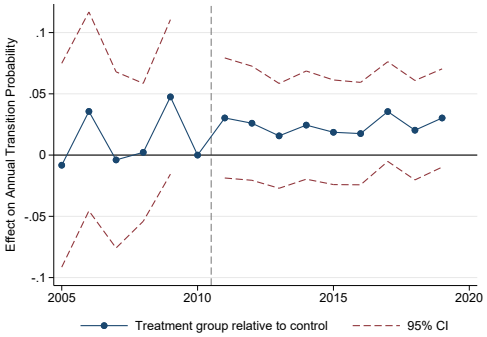
in this scenario. The benefits of GAIN are heavily skewed toward drugs with low pre-GAIN total exclusivities, such that the average percent increase in exclusivity in the simulation exercise is +58%. As 48% of the drugs in our treatment group received GAIN exclusivity after 2012, we calculate that the GAIN Act increased total exclusivity by $58\% \times 48\% = 28\%$.

Combining these estimates with our baseline DID estimates from Table 2 yields revenue elasticities in the range of $0.94/0.36 = 2.6$ to $0.94/0.28 = 3.4$.

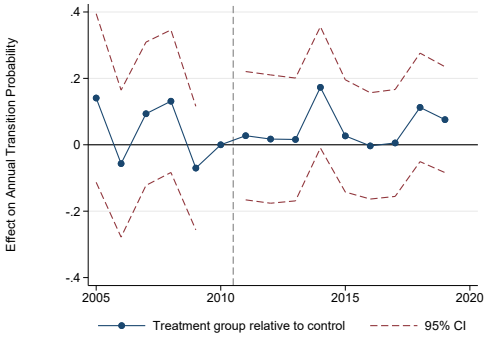
C Event studies for all adjacent phases

Appendix Figure A1. Event studies by phase

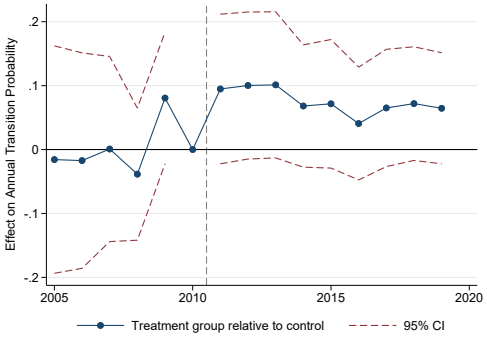
(a) Early stage to phase 1



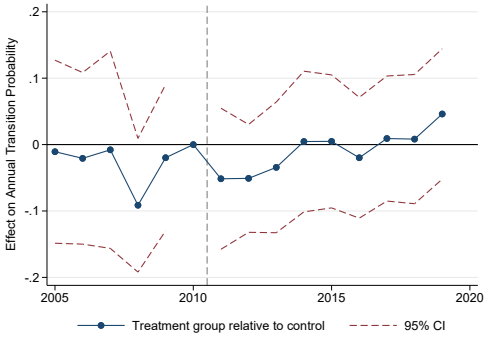
(b) Phase 1 to phase 2



(c) Phase 2 to phase 3



(d) Phase 3 to launch



Notes: Figure shows the baseline event study specification for adjacent phase transitions. The Y-axis shows the probability of advancing to a later phase for drug candidates with treated indications (antibiotics) versus control indications (antivirals and antibacterial vaccines). The dashed gray vertical line indicates the timing of the GAIN Act; 2011 is the first post-period year. The control group average has been normalized to have the same pre-period mean as the treatment group.

D Pooled difference-in-differences for adjacent phases

Table A1 below shows the full set of pooled difference-in-difference results for all adjacent phase pairs estimated through Equation (3). Effects are concentrated on the phase 2 to 3 transition. With drug fixed effects (Column 4), we also see modest effects for phase 1 to 2.

Appendix Table A1. Pooled Difference-in-Differences Results for Adjacent Phase Transitions

	(1) Baseline	(2) Dose X Treatment	(3) Biologic X Treatment	(4) Drug FE
Panel (a): Phase 0 to phase 1				
Post X Treatment	0.010	0.003	0.003	-0.021+
	(0.012)	(0.012)	(0.012)	(0.011)
Pre-period mean				
Treatment	0.070	0.070	0.070	0.070
Control	0.041	0.041	0.041	0.041
N unique drugs	3,352	3,352	3,352	2,802
N	17,163	17,163	17,163	16,613
Panel (b): Phase 1 to phase 2				
Post X Treatment	0.017	0.007	0.008	0.125*
	(0.043)	(0.043)	(0.043)	(0.058)
Pre-period mean				
Treatment	0.225	0.225	0.225	0.225
Control	0.146	0.146	0.146	0.146
N unique drugs	1,027	1,027	1,027	806
N	4,005	4,005	4,005	3,784
Panel (c): Phase 2 to phase 3				
Post X Treatment	0.069*	0.060*	0.064*	0.100**
	(0.028)	(0.028)	(0.028)	(0.038)
Pre-period mean				
Treatment	0.068	0.068	0.068	0.068
Control	0.077	0.077	0.077	0.077
N unique drugs	741	741	741	633
N	3,897	3,897	3,897	3,789
Phase (d): Phase 3 to launch				
Post X Treatment	0.015	0.018	0.020	-0.008
	(0.021)	(0.021)	(0.021)	(0.022)
Pre-period mean				
Treatment	0.065	0.065	0.065	0.065
Control	0.054	0.054	0.054	0.054
N unique drugs	504	504	504	455
N	3,177	3,177	3,177	3,128

$^{+}p < 0.1,^{*}p < 0.05,^{**}p < 0.01$

NOTES: Table shows pooled event study estimates of yearly phase transition rates from Equation (3), separately for each pair of adjacent phases. The coefficient of interest, Post X Treatment, reflects the effect of the GAIN Act on the transition rate of antibiotics versus a control group comprised of antivirals and vaccines. Column 1 shows our baseline specification. Column 2 adds dosage form indicators and interactions with the treatment group. Column 3 adds a biologic indicator and interaction with the treatment group. Column 4 controls for drug fixed effects.

E Alternative specification: clinical trial counts

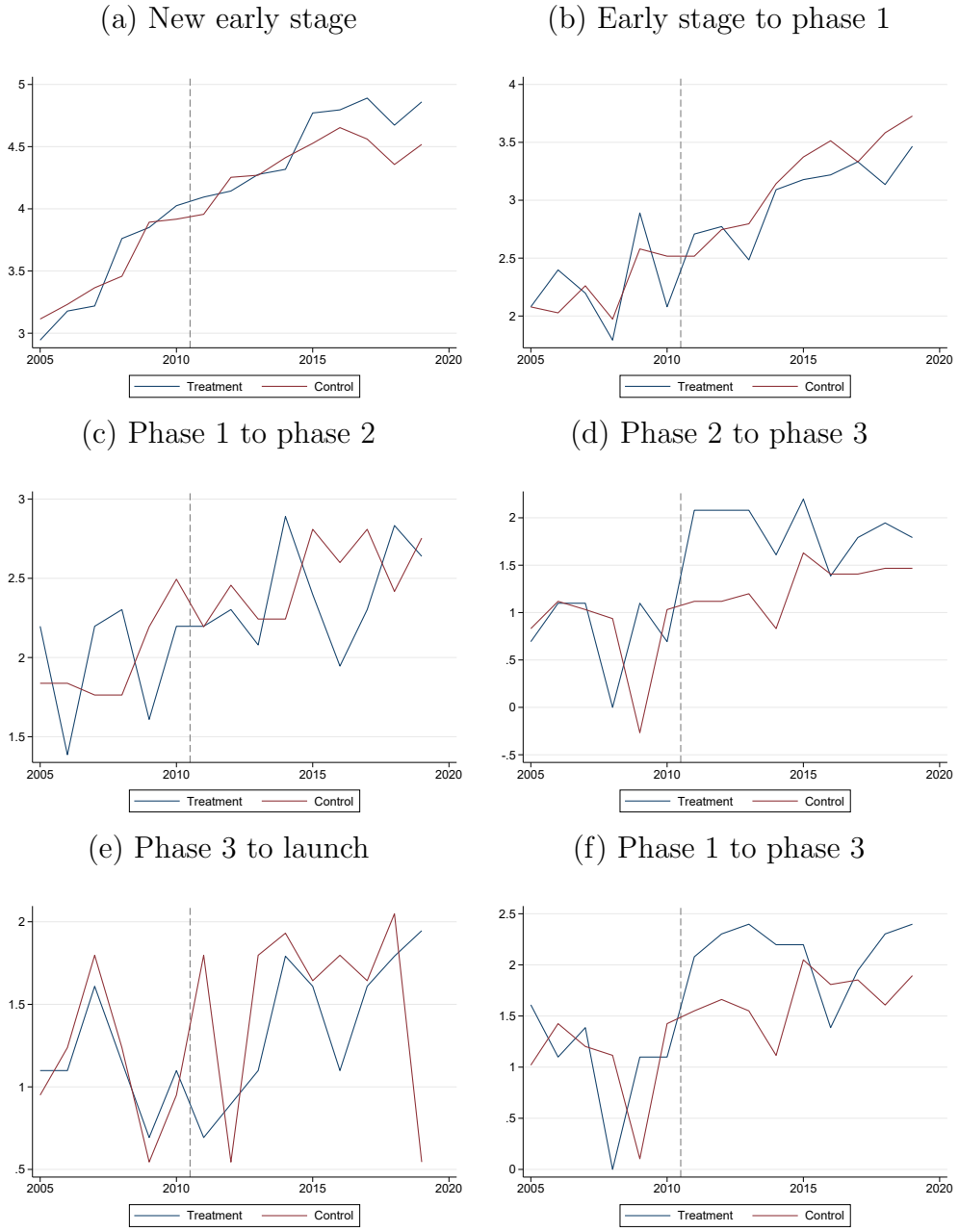
Our main text specification examines clinical trial transition rates at the individual drug candidate level. This accounts for the size of the "at-risk" pool of drug candidates at a given phase in addition to the number of candidates that advance to the next phase. Alternatively, one could simply compare the (log) total number of trial initiations at each phase, where initiation counts are log transformed to account for the size difference between the treatment and control groups. This method complicates the inclusion of control variables (as we explain further below), but it allows us to examine the number of new pre-clinical milestones (i.e., early-stage or "phase 0" drug candidates). Unlike later phases, we cannot use the transition rate framework for phase 0 candidates because we do not observe the pool of drug candidates that are "at risk" of starting phase 0 trials.

Figure A2 shows comparisons of the log number of initiations in the treatment versus control group. The pre-period average of the control group is normalized to equal the pre-period average of the treatment group for easier comparison of post-GAIN effects. There is some evidence for an increase in new early-stage milestones (i.e., pre-clinical studies) starting in 2015 (Panel a). We find that most phase transitions do not have visually appreciable effects (e.g., Early stage to phase 1, phase 1 to phase 2, or phase 3 to Launch). Consistent with the main text results, there are potential increases in trials between phase 2 and 3 (Panel d) and between phase 1 and 3 (Panel f).

Figure A3 shows differences in the log number of initiations in the treatment versus control group, with 95% confidence intervals obtained via bootstrap. As in Appendix Figure A2, no controls are included (adding controls is generally not feasible as this results in many group-year cells with zero initiations, for which the log is not defined).

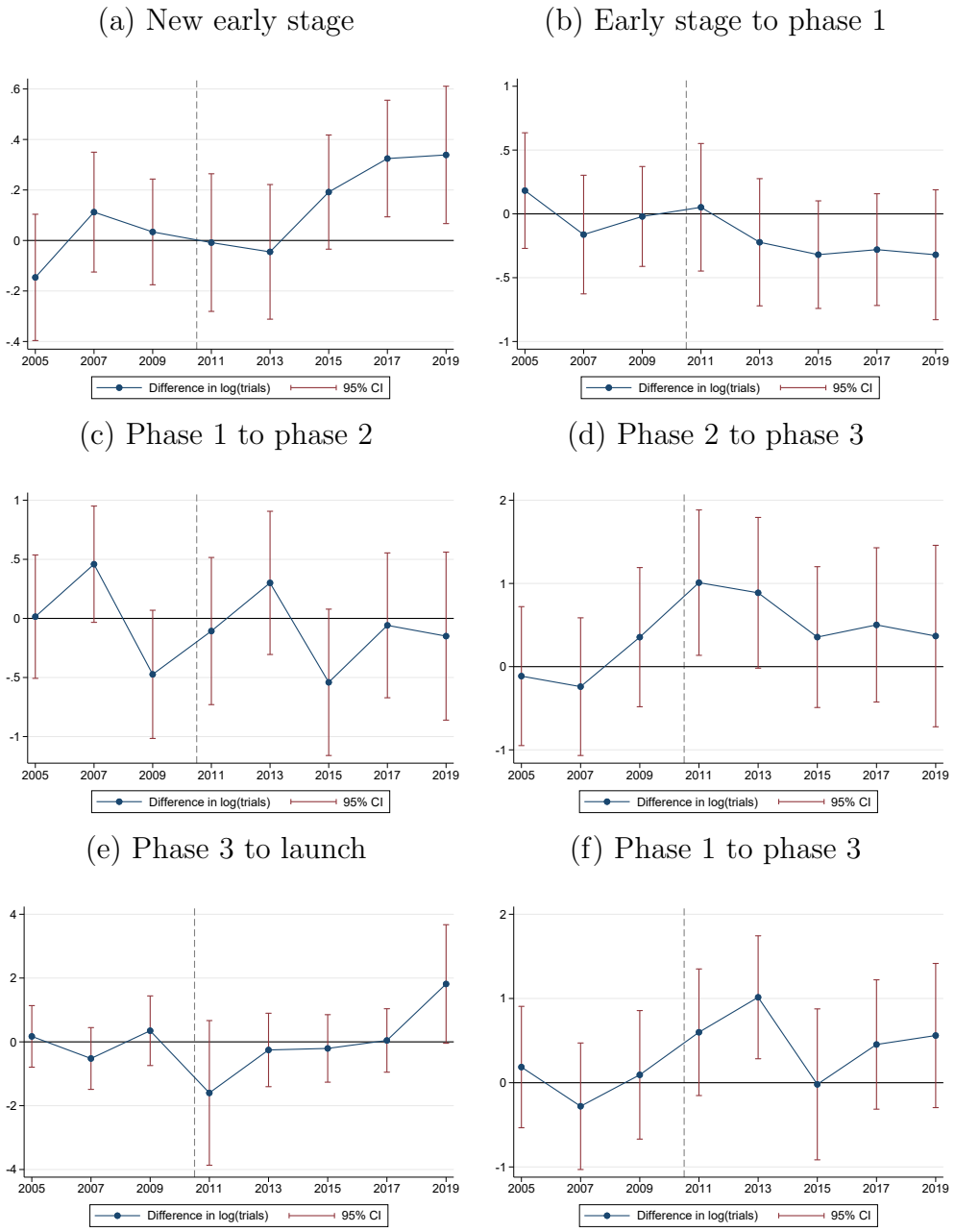
First, we see that pre-period trends in Figure A3 are quite flat, lending support to the difference-in-differences parallel trends assumption. We find evidence for a delayed uptick in new early-stage (phase 0) milestones, as discussed in the main text. We find some evidence for increased phase 3 trials (Panels d and f) in the first 4 years after the GAIN Act (2011-2014). The temporary nature of this estimate is not surprising, as an increase in phase 3 trials (without a commensurate increase at early stages) depletes the size of the "at-risk" pool of phase 2 trials that have not yet started phase 3. This highlights a benefit of our transition rate analysis in the main text, which adjusts for the denominator of at-risk phase 2 trials. The other phases do not show significant increases in log trials.

Appendix Figure A2. Number of Log Clinical Trial Initiations by Phase



Notes: Figure shows the number of first trials for treatment versus control groups of drug candidates at each phase transition. The Y-axis shows the log number of drug candidates whose first milestone of a given phase occurred in that year. The dashed gray vertical line indicates the timing of the GAIN Act, where we define 2011 as the first post-period year. The control group average has been normalized to have the same pre-period mean as the treatment group. Panel (a) shows the (log) number of drug candidates with an early-stage (pre-clinical) or later trial. Panel (b) shows the log number of phase 1 (or higher) trials, among drugs that are observed in pre-clinical studies. Panel (c) shows the log number of phase 2 (or higher) trials, among drugs that are observed in phase 1 studies. Panels (d) and (e) show analogous results for phase 3 and launch, respectively. Panel (f) shows the log number of phase 3 trials (or launches) among drugs that are observed to have a phase 1 or 2 trial earlier.

Appendix Figure A3. Differences in Log Clinical Trial Initiations by Phase



Notes: Figure shows differences in the log number of trials for treatment versus control groups of drug candidates at each phase transition. Data are binned into two-year bins except for 2019, which contains one year. Error bars represent 95% confidence intervals, obtained via 200 Bayesian bootstrap iterations. The dashed gray vertical line indicates the timing of the GAIN Act, where we define 2011 as the first post-period year. The control group average has been normalized to have the same pre-period mean as the treatment group for each bootstrap iteration.

Table A2 below quantifies the log-initiation results by pooling data in the pre- and post-periods. Panel (a) quantifies the results shown above in Appendix Figure A3. Panel (b) adjusts for cohort X treatment group interactions by summing up initiations separately by cohort (defined as the first year a drug appears in our dataset at any phase). Overall, the results from this alternative specification are consistent with our baseline results in the main text, but with less statistical power. Column 6 of Panel (a) shows that, among all drugs reaching phase 1 or 2, the GAIN Act was associated with an approximately 40% increase in phase 3 (or higher) trial initiations. After adjusting for cohort-treatment group interactions (Panel b), the magnitude of the phase 1-3 estimate increases and is statistically significant (P = 0.019). Consistent with our analysis in the main text, the largest effects were driven by the phase 2-3 transition, with smaller, null effects for other phases.

In general, we view these log-initiation results as broadly consistent with our main text results on transition rates. The log-initiation specifications are less well-powered for two reasons. First, the results in Panels (b) – (f) do not account for the denominator: the number of drugs "at-risk" that are available to transition to the next phase in each year. Second, the log-initiation specifications do not allow for adding detailed controls, since this requires dividing the data into smaller bins. Because the outcome is the log of the number of initiations, bins that happen to contain zero initiations are undefined.

Appendix Table A2. Aggregate DID Estimates for the ln(initiation) Specification

	(1) Phase 0	(2) Phase 0-1	(3) Phase 1-2	(4) Phase 2-3	(5) Phase 3-L	(6) Phase 1-3
Panel (a) Without controls						
Treatment X Post GAIN	0.143	-0.215	-0.099	0.614+	-0.015	0.442
SE	(0.096)	(0.162)	(0.208)	(0.318)	(0.374)	(0.292)
Panel (b) With adjustment for cohort X treatment group						
Treatment X Post GAIN	-1.227	0.092	-0.096	0.978+	-0.327	1.230*
SE	(0.933)	(0.453)	(0.354)	(0.580)	(0.693)	(0.523)
N (drug-years)	7,876	17,124	4,002	3,895	3,177	7,398

$^{+}p < 0.1,^{*}p < 0.05,^{**}p < 0.01$

NOTES: Table shows results from difference-in-differences comparisons regressing log trials on a treatment group indicator, post GAIN Act (year $\geq$ 2011) indicator, and their interaction. Reported effects are for the treatment X Post-GAIN interaction term. Panel (a) does not include any controls and thus reflects the graphical results shown in Appendix Figure A3, whereas Panel (b) controls for cohort (the first year a drug candidate is observed in the dataset at any phase) and treatment group interactions. Standard errors are obtained via bootstrap at the drug candidate level with 200 re-sampling draws; Panel (b) uses a Bayesian bootstrap to avoid cases where re-sampling draws contain no observations for particular cells.

F Pooled estimates for analyses of patenting activity

The effects of the GAIN Act on patenting varied by patent and owner types. Table A3 shows that how patenting effects varied by patent type, with the largest increases among diagnostic, combination, formulation, new use, and process patents. Product patents exhibited a significant, but smaller, increase. As a placebo test, we show that patents for biologics (which were not eligible for GAIN Act exclusivity) did not significantly increase.

Appendix Table A3. Effects on log Patents, by Patent Type

Patent type	(1) Aggregate	(2) Product	(3) Diagnostic	(4) Ancillary	(5) Biologics
Treatment X Post GAIN	0.387**	0.117**	0.392**	0.231**	0.123
SE	(0.018)	(0.035)	(0.050)	(0.028)	(0.079)
N (patents)	53,746	15,524	8,507	30,267	3,653

$^{+}p < 0.1,^{*}p < 0.05,^{**}p < 0.01$

NOTES: Table shows results comparing treatment and control groups before vs. after the GAIN Act, separately by patent type. The Ancillary patent type is defined to include combination, formulation, new use, and process patents. Biologics were not covered by the GAIN Act and can therefore be considered a negative control or placebo outcome.

Table A4 shows how effects varied by owner type, comparing estimates for all companies (Column 1), for-profit pharmaceutical firms (Column 2), and non-pharmaceutical entities (Column 3). Non-pharmaceutical companies exhibited larger relative increases, but account for fewer patents overall. Table A5 shows that effects on product patents were primarily driven by non-pharmaceutical firms.

Appendix Table A4. Effects on log Patents, by Owner Type

Patent owner type	(1) Aggregate	(2) Pharmaceutical	(3) Non-pharmaceutical
Treatment X Post GAIN	0.387**	0.341**	0.464**
SE	(0.018)	(0.023)	(0.035)
N (patents)	53,746	32,328	17,619

$^{+}p < 0.1,^{*}p < 0.05,^{**}p < 0.01$

NOTES: Table shows results comparing treatment and control groups before vs. after the GAIN Act, separately by the type of patent owner. Non-pharmaceutical owners are primarily comprised of universities, hospitals, and non-profit institutes.

Appendix Table A5. Effects on log Product Patents, by Owner Type

	(1) Pharmaceutical	(2) Non-pharmaceutical
Treatment X Post GAIN	-0.048	0.346**
	(0.048)	(0.064)
N (patents)	10,371	5,241

$^{+}p < 0.1,^{*}p < 0.05,^{**}p < 0.01$

NOTES: Table shows results comparing treatment and control groups before vs. after the GAIN Act, limiting to product patents, separately by the type of patent owner. Non-pharmaceutical owners are primarily comprised of universities, hospitals, and non-profit institutes.

G Impact of the Inflation Reduction Act on innovation

Using the FDA Orange Book data, we compute that an average of 83.5 small-molecule branded drugs were approved each year from 2005 through 2019. The Inflation Reduction Act (IRA) allows the government to negotiate Medicare prices for up to 20 additional drugs each year, with the set of price-negotiated drugs expanding over time. However, due to eligibility criteria, negotiations may apply to fewer than 20 drugs.[36]. Estimates by Vogel et al. (2024) suggest that on average, negotiation may apply to 7.4 new drugs per year. In the long run, this implies that a randomly chosen new drug has a $7.4/83.5 = 8.9\%$ probability of being subject to price negotiation.

To model the effects of the IRA as a cap on market exclusivity, we use the same distribution of pre-GAIN total exclusivity as in Appendix Section B. Because the IRA allows for price negotiation to occur after 9 years of market exclusivity for small molecule drugs, we simulate a 9-year exclusivity cap, which reduces the total exclusivity of affected drugs (those who would have had > 9 years of exclusivity) by an average of 5.2 years, or an average percent decline of 33%. We make the conservative assumption that all drugs are equally likely to be negotiated, implying an expected percent decline in total exclusivity of $8.9\% \times 33\% = 3.0\%$. In Appendix Section B, we computed that the GAIN Act increased total exclusivity by 28%. Hence, we infer that the relative effect of the IRA on expected exclusivity is 3.0% / 28% = 0.107.

The next step requires estimating the reduction in a drug's US revenue that results from negotiation, compared to the effect of generic entry, which we conservatively assume is a 100% reduction in revenue. This is a function of (1) the revenue decrease negotiated by Medicare and (2) the share of a drug's US revenue that derives from Medicare. For (1), Hernandez et al. (2024) use the actual, newly negotiated maximum fair prices for the first 10 drugs subject to the IRA and find a 22% savings to Medicare on these drugs. For (2), we use the facts that Medicare accounted for \$256 billion (42%) of the \$603 billion in total US drug spending in 2021 through Part B and Part D (Parasrampuria and Murphy, 2022; Kaiser Family Foundation, 2023). Thus, we estimate that Medicare price negotiation results in a $0.22 \times 0.42 = 9.2\%$ decrease in US revenues.

Recall our main estimates that the GAIN Act increased antibiotic preclinical studies by 33% by the end of our sample window and phase 3 trials by 94%, for a total effect of 127%.[37] From this, we can infer that the IRA will lower drug innovation by approximately

[36]For more details on the drug negotiation program, see https://www.kff.org/medicare/issue-brief/faqs-about-the-inflation-reduction-acts-medicare-drug-price-negotiation-program/

[37]We assume that the increase we estimate for in antibiotic patenting is subsumed in the effect on preclinical studies, rather than additive.

$0.107 \times 9.2\% \times 127\% = 1.3\%$, or a loss of $1.3\% \times 83.5 = 1.05$ drugs per year.

Effects may be much larger for drugs that are mainly used for elderly patients on Medicare. The maximum impact of negotiation on US revenues, assuming a 100% Medicare share, would be 22%, implying a larger $0.107 \times 22\% \times 127\% = 3.0\%$ decline.

Our calculation may be too conservative for several reasons. First and foremost, the IRA is specifically targeted towards drugs with the highest Medicare revenues, whereas our estimates for the GAIN Act pertain to the average antibiotic. The targeting of the IRA toward drugs that (ex-post) generate the highest revenues magnifies its effect on total profits for the industry as a whole, compared to policies like the GAIN Act that are not targeted based on ex-post revenues. Moreover, the IRA will also likely induce shifts in innovation away from drugs treating conditions of the elderly. Second, we did not include our estimated 47% increase in antibiotic patenting. Although some of this effect is likely subsumed in the 33% increase in preclinical studies observed by 2019, it may also lead to greater extensive-margin increases in innovation in the long-run. Lastly, antibiotics may be less responsive to exclusivity changes compared to other drugs, especially if new antibiotics are mostly held in reserve and annual revenues are low at baseline. This would imply larger innovation effects of the IRA for non-antibiotics.

Antibiotics may be more responsive to exclusivity changes compared to other drugs. For instance, profitability may be a more significant barrier for antibiotic innovation, compared to other areas where scientific considerations constitute the primary barriers. If this is the case, then our estimate for IRA innovation effects would be too large.

For comparison, the Congressional Budget Office estimated that the IRA would reduce drug innovation by about 1.1% over the next 30 years.[38]

While this calculation involves many assumptions, our hope is that it provides a useful framework for translating our results to other settings. One could modify each of the above assumptions to reach alternative estimates of the effect of the IRA. For example, if one believed that the innovation elasticity for antibiotics was two times *larger* than that of drugs affected by the IRA, this would cut the inferred IRA innovation effect in half.

[38] This is based on a decline of 15 drugs out of 1300 drugs approved over the next 30 years. See p. 5 of https://www.cbo.gov/system/files/2022-07/senSubtitle1_Finance.pdf

H Phase Advancement Probabilities

Table A6 below reports phase-specific advancement probabilities, computed for each phase k, k' pair as the share of drug candidates observed to reach phase k by 2018 that are also observed to have a subsequent phase k' trial.[39]

Appendix Table A6. Phase Advancement Probabilities

	Treatment	Control	Total
Pre-clinical to phase 1			
N Pre-clinical	1,096	2,169	3,265
Share advancing to phase 1	26%	31%	29%
Phase 1 to 2			
N phase 1	292	770	1,062
Share advancing to phase 2	61%	51%	54%
Phase 2 to 3			
N phase 2	240	576	816
Share advancing to phase 3	50%	46%	47%
Phase 3 to US launch			
N phase 3	174	396	570
Share launched	51%	32%	38%
Phase 1 to 3			
N phase 1 or 2	368	1,008	1,376
Share advancing to phase 3	40%	33%	35%

NOTES: For the "Phase 1 to 3" transition, reported probabilities are calculated using the sample of drugs that we observe in either phase 1 or phase 2 clinical trials by 2018. Drugs with both phase 1 and phase 2 activity observed by 2018 only enter the sample once (they are not "double counted"). Sample sizes reflect the number of unique drug candidates (compounds), aggregating over all of the indications of each drug.

On average across both treatment and control, 29% of drugs observed in pre-clinical studies progress to at least phase 1 trials, 54% of phase 1 drugs progress to at least phase 2, 47% of phase 2 drugs progress to phase 3, and 38% of phase 3 drugs are launched in the US. We can see that antibiotics have somewhat lower rates of progression from pre-clinical to phase 1. However, conditional on reaching phase 1, a similar share of antibiotics progress to phase 2 (61% versus 51% of controls) and from phase 2 to phase 3 (50 versus 46%). Conditional on reaching phase 3, a larger share of antibiotics is launched (51% versus 32%).

[39] We require that phase k begins by 2018 because our data only extend through 2021. This allows each drug candidate at least three years to advance to phase k'

Overall success rates: To get an overall launch rate using the full sample of drugs (as opposed to just the drugs observed going from pre-clinical to launch), we use the individual transition probabilities and assume independent transition probabilities. Thus, the overall launch rate of a pre-clinical antibiotic is $26\% \times 61\% \times 50\% \times 51\% = 4.0\%$ and the overall launch rate of a pre-clinical antiviral drug is 2.3%. An antibiotic that reaches phase 1 has a 16% probability of being launched, compared to an 8% probability for control group drugs, with most of the difference accounted for by a smaller launch probability for control drugs.

Comparisons of success rates to the literature: Table A7 below provides estimates of conditional transition probabilities in the existing literature that are specific to antibiotics or infectious disease drugs more broadly.

Our values are roughly similar for the transitions from phase 1 through phase 3 trials. Our estimates of transitions between pre-clinical to phase 1 trials are higher. One possible reason is that reporting of pre-clinical activity (which tend to be less publicized compared to clinical trials and drug approvals) likely exhibits the most variance across data sources and that Cortellis might be selectively reporting pre-clinical activity that leads to phase 1 trials.

Relative to the literature (Appendix Table A7), the drugs in our sample are less likely to launch in the US after initiating phase 3 trials. One reason for the discrepancy may be the types of firms and drug products included in our sample compared to the literature. Our sample of antibiotics, antivirals, and vaccines may be less commercializable than the average infectious disease drug in the literature. In addition, our success probabilities only count US launches associated with a treatment or control indication: drugs that have a treatment or control indication in phase 3, but are launched under other indications, are not coded as successes in our analysis. Reassuringly, we are able to closely match overall success rates from Aryal et al. (2023), which also uses the Cortellis dataset.

Appendix Table A7. Probability of Success Estimates from the Literature

Paper	Pre-clinical to Phase 1	Phase 1 to 2	Phase 2 to 3	Phase 3 to Launch [a]
Dowden and Munro (2019)		0.59	0.38	0.71
Wong et al. (2019)		0.70	0.58	0.75
Stephens (2015)	0.09	0.33	0.75	0.64
Hay et al. (2014)		0.67	0.46	0.63
Thomas et al. (2016)		0.70	0.43	0.73
Payne et al. (2007)	0.26	0.25	0.5	0.50
DiMasi et al. (2010)		0.58	0.52	0.79
DiMasi et al. (2020)[b]		0.76	0.53	0.78
Range across above papers	0.09–0.26	0.25–0.76	0.35–0.75	0.50–0.79
Average in our sample	0.29	0.54	0.47	0.38
Aryal et al. (2023)[c,d]	Pre-clinical to launch for infectious diseases: 0.16			

NOTES: Table shows transition probability estimates from the literature. Other than Aryal et al. (2023), the included papers satisfy the following criteria: (1) cited in the literature review of a model of antibiotic value developed by researchers at Boston University (Outterson, 2021) (2) report estimates based on authors' own analysis of data (3) published as institutional report or in peer-reviewed academic journals.
[a] Most studies usually decompose the "phase 3 to launch" transition into at least two discrete steps, with intermediate milestones of FDA application submission and FDA approval. We have collapsed these steps into one transition, and the values in this column are found by assuming independence and multiplying all relevant conditional transition probabilities.
[b] Study uses regulatory approval in other countries around the world, not just FDA approval and launch in the US market.
[c] Study also uses *Cortellis* as a data source.
[d] Conditional transitional probabilities calculated at the drug-indication level. Parameters from all other papers calculated at the drug-level or only for a drug's lead indication.

Other studies that use the *Cortellis* data: Aryal et al. (2023) also uses *Cortellis* data to estimate conditional transition probabilities. The authors use a sample constructed at the firm-drug-indication-level and report a success rate of 72.3% in transitioning from phase 3 to FDA application and success rate of 89.0% in transitioning from FDA application to approval, so we infer a phase 3 to approval success rate of $72.3\% \times 89.0\% = 64\%$. Specific to infectious disease products, the authors report that 16% of drugs with pre-clinical activity eventually go to market. Using a drug-indication version of our sample, we find a conditional transition probability of 62% between phase 3 and launch and 19% between preclinical and US launch, which closely match the other *Cortellis* estimates in Aryal et al. (2023).